

Reconsidering Misinformation within
Adaptive Socio-Technical Communication
Environments:
A Conceptual Exploration

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**Australian
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Disclaimer

I hereby declare that the work undertaken in this thesis has been conducted by me alone, except where indicated in the text. I conducted this work between January 2026 and June 2026, during which period I was a Master student at the Australian National University. This thesis, in whole or any part of it, has not been submitted to this or any other university for a degree.

A handwritten signature in black ink, consisting of the Chinese characters '郑岳' (Zheng Yue) in a cursive style.

Yue Zheng
June 2026

Abstract

Contemporary communication environments are characterised by unprecedented volumes of information, diverse sources of content, and rapidly evolving forms of communication. Under these conditions, identifying and interpreting misinformation has become increasingly challenging, raising broader questions about how misinformation can be understood and observed.

Through a review of the literature, this paper examines misinformation through multiple analytical perspectives, including information diffusion, human behaviour, platform dynamics, and computational detection. The review shows that many existing approaches rely on indicators and classifications that can become unstable as communication environments change. This raises a broader challenge: how can misinformation continue to be meaningfully observed under changing conditions?

To address this question, the study proposes a cybernetic framework for misinformation observation. Unlike conventional detection approaches that focus primarily on determining whether information is true or false, the framework treats observation as an adaptive process linking computational representations, human interpretation, and changing communication environments. Rather than producing definitive classifications, the framework is designed to identify conditions under which existing representations may no longer adequately reflect reality. Under such conditions, early warning signals can prompt human intervention and the re-evaluation of ground truth, enabling observational systems to adapt as communication environments evolve. In this way, the framework provides a potential bridge between computational models and changing real-world conditions, shifting attention from detection towards adaptive observation and early warning.

Keywords: misinformation; information diffusion; human information interaction; platform dynamics; misinformation detection; cybernetics; early warning

AI Usage Statement

This thesis made use of generative artificial intelligence (AI) tools, including ChatGPT, as writing and research support tools during the preparation of the manuscript. AI-assisted support was used in the following areas:

1. Translation of content from the author's native language into English: AI tools were used to assist in translating draft notes and text from the author's native language (Chinese) into academic English. In cases where the author had already determined the intended meaning but was uncertain about the most appropriate English terminology or expression, AI tools were used to suggest suitable vocabulary and alternative sentence constructions. The underlying ideas and intended meanings originated from the author.
2. Language editing, grammar correction, and stylistic refinement: AI tools were used to review draft paragraphs written by the author and provide suggestions for grammar correction, language refinement, sentence restructuring, and improvements in readability. These interactions focused on improving the clarity and fluency of expression rather than generating original arguments or content.
3. Recommendations regarding structure, organisation, logical flow, and content reduction: AI tools were used during the revision stage after the completion of a full draft of the thesis. The initial manuscript exceeded the target word limit (approximately 26,000 words) and therefore required substantial consolidation and reduction. AI tools were used to help identify repetitive arguments, overlapping discussions across chapters, and sections that could potentially be merged or shortened. Following these revisions, AI tools were also used to review whether the overall logical progression and chapter-level coherence remained intact after restructuring and content reduction.
4. General academic writing feedback and discussion of alternative ways to present ideas: As a first-time author writing a thesis entirely in English, the author used AI tools to seek advice on academic writing conventions and manuscript presentation. This included discussions of paragraph organisation, argument presentation, the use of transitions and linking expressions, and common practices in academic writing. AI tools were used as a learning and feedback resource to support the author's development of academic writing skills rather than as a source of original arguments or research content.
5. Minor supplementary literature recommendations: During the development of Chapters 3–6, AI tools were occasionally used to discuss potential gaps in the logical progression of arguments and identify areas where additional supporting discussion might be beneficial. In some cases, AI tools suggested supplementary literature that could potentially address these gaps or strengthen connections between sections. All recommended sources were independently

located, verified, evaluated, and selected by the author before inclusion in the thesis.

6. Exploratory discussions and early-stage brainstorming: During the exploratory phase of the project (January–February 2026), AI tools were used to support brainstorming and the exploration of different research perspectives within the misinformation literature. These discussions assisted the author in becoming familiar with major research areas, identifying potential directions for further investigation, and comparing alternative ways of framing the research topic. The selection of the final research focus and analytical approach remained the responsibility of the author.

AI tools were not involved in the following core research activities:

1. Formulation of the research questions and overall research objectives;
2. Design of conceptual models and theoretical frameworks;
3. Development of the analytical structure and overall argument of the thesis;
4. Identification, selection, and prioritisation of core literature;
5. Verification, interpretation, and placement of literature, citations, and supporting evidence;
6. Formulation of chapter-level arguments, analytical insights, and conclusions;
7. Final interpretation, evaluation, and synthesis of findings, including the formulation of the overall contribution, proposed framework, and future directions of the thesis;
8. Design of figures and tables;

All AI-assisted outputs were critically reviewed, revised, and verified by the author. The author takes full responsibility for the accuracy, interpretation, arguments, and conclusions presented in this thesis.

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Chapter 1

Introduction

1.1 A Glimpse from History to the Present

“The Corsican Monster has landed.”

“The Usurper entered Grenoble.”

“Napoleon approaches Paris.”

“His Imperial Majesty has arrived safely in France.”

These reports were published during Napoleon’s return to France in 1815.

Three of these descriptions referred to the same person.

For ordinary readers in France at that time, which version of Napoleon should be believed? Was Napoleon a monster, a usurper, or once again an emperor? More importantly, if different reports each appeared persuasive under different circumstances, which of them should be understood as misinformation—and were any of them necessarily false?

Although the event itself remained unchanged, its meaning appeared to shift as descriptions, audiences, political conditions, and interpretive environments evolved. For people living in France in 1815, understanding and distinguishing competing descriptions was already far from straightforward.

This challenge remains visible today.

Unlike readers in France in 1815, contemporary audiences no longer encounter only a handful of competing reports but increasingly fragmented and continuously evolving communication environments.

According to the Reuters Institute Digital News Report 2025, news consumption across online platforms continues to fragment, with six online networks now each reaching more than 10% weekly for news consumption compared with only two a decade earlier. At the same time, social video news consumption increased from 52% in 2020 to 65% in 2025, further accelerating the circulation and accessibility of information. Yet greater informational exposure does not necessarily guarantee greater understanding. More than half (58%) of respondents reported concerns about their ability to distinguish what is true from what is false online, while four in ten (40%) reported sometimes or often avoiding the news altogether ([Newman 2025](#)).

Similar concerns are reflected in Australia’s broader information environment. A recent report by the Australian Communications and Media Authority found that 74% of Australian adults expressed concern about online misinformation—the highest level reported globally—and noted that declining trust in mainstream news, increasing reliance on digital platforms, and gaps in media literacy are making it

increasingly difficult for individuals to identify and respond to misinformation ([Australian Communications and Media Authority 2025](#)).

Taken together, these findings suggest that contemporary audiences are increasingly exposed to large volumes of information while remaining uncertain about how information should be interpreted and trusted. Distinguishing between true and false information may therefore become increasingly demanding, contributing not only to uncertainty but also to disengagement from news and public information environments.

Under such conditions, misinformation may become increasingly difficult to understand and observe through conventional assumptions alone. Revisiting how misinformation is understood—and how people engage with observing information—may therefore provide an opportunity to reconsider how meaningful understanding can be maintained under contemporary communication environments.

1.2 Research Questions and Objectives

The discussions above suggest that contemporary challenges surrounding misinformation may extend beyond identifying isolated false information and increasingly concern how misinformation continues to be understood and observed under changing communication environments.

Rather than treating misinformation as a fixed object waiting to be detected, this thesis approaches misinformation as a phenomenon whose visibility and interpretation may emerge differently across changing communication conditions.

This perspective gives rise to the following research question:

RQ: How can misinformation continue to be understood and observed within adaptive socio-technical communication environments?

To address this question, two sub-questions guide the development of this thesis.

SubRQ1: How do existing approaches construct observational perspectives for understanding misinformation?

This question examines how different traditions of misinformation research construct particular ways of making misinformation visible and understandable. Rather than evaluating approaches through predictive performance alone, attention is directed towards the assumptions, boundaries, and observational perspectives embedded within existing literature.

SubRQ2: What challenges emerge when computational observation encounters adaptive communication environments, and how may observation remain meaningful under such conditions?

This question explores how changing communication environments may challenge existing observational approaches and investigates possibilities for maintaining meaningful observation under conditions of adaptation, feedback, and environmental change.

Building upon these questions, this thesis pursues three objectives:

- To examine how existing approaches construct ways of understanding and observing misinformation;
- To identify limitations that emerge under changing communication environments;
- To develop a conceptual framework for maintaining meaningful observation under adaptive conditions.

1.3 Thesis Structure

To address the central research question of how misinformation may continue to be understood and observed within adaptive socio-technical communication environments, this thesis progressively approaches the problem through a series of smaller questions.

Table [1.1](#) summarises the role of each chapter and illustrates how different parts of the thesis contribute towards answering the sub-questions and, ultimately, the broader research question.

Together, these chapters progressively contribute towards answering the central research question.

Table 1.1: Structure of the Thesis and Relationship to Research Questions

Chapter	Guiding Question	Related SubRQ	Contribution
C2	What is misinformation (from a systems perspective)?	SubRQ1	Reviews dominant conceptualisations of misinformation and clarifies the systems-oriented understanding of misinformation adopted in this thesis.
C3	How does misinformation spread across communication systems?	SubRQ1	Explores diffusion processes and identifies distinctive diffusion characteristics through which misinformation becomes observable within communication systems.
C4	How do human behaviours shape engagement with misinformation?	SubRQ1	Examines psychological, cognitive, and social mechanisms underlying human–information interaction and identifies recurring interaction patterns through which misinformation becomes observable.
C5	How do platform environments shape misinformation diffusion and interaction?	SubRQ1	Examines how platform architectures influence visibility, amplification, and governance processes, shaping the conditions under which misinformation diffusion and human–information interactions occur.
C6	How is misinformation operationalised and detected within contemporary communication systems?	SubRQ2	Reviews dominant detection approaches and examines how misinformation is operationalised through content, diffusion, and behavioural indicators, while identifying limitations under adaptive communication environments.
C7	How may misinformation continue to be meaningfully observed?	SubRQ2	Proposes a conceptual framework for supporting adaptive misinformation observation through early warning and human-guided updating of observational systems.

Chapter 2

Understanding Misinformation as a System

What is misinformation? How can misinformation be understood? Why is misinformation different from truthful information? Can information that is considered misinformation in one context remain misinformation in another?

Although these questions appear straightforward, answering them is often far from simple. Contemporary communication environments contain a diverse range of misleading information, making it increasingly difficult to determine what should be considered misinformation and how it should be understood.

To address these questions, this chapter examines how misinformation has been understood within existing literature and introduces a system-oriented perspective for understanding misinformation in contemporary communication environments, thereby establishing a conceptual foundation for the subsequent chapters to observe misinformation from different perspective.

2.1 Understanding Misinformation

The rapid expansion of digital communication platforms has fundamentally transformed how information is produced, distributed, and consumed. Social media platforms enable information to circulate rapidly across large audiences, often bypassing traditional gatekeeping institutions such as professional journalism. While these platforms facilitate unprecedented access to information, they also create conditions that enable the widespread circulation of misleading or false content. As a result, misinformation and disinformation have emerged as major concerns in contemporary societies, particularly in relation to democratic processes, public health communication, and the broader information ecosystem ([Lewandowsky et al. 2017](#), [Lazer et al. 2018](#)).

Philosophical discussions of information provide an important foundation for understanding the nature of misleading information. In early work on semantic information, [Floridi \(2004\)](#) argues that information should be understood as meaningful data that carries truth-value conditions. From this perspective, false or misleading information represents a distortion of informational content within communication systems. Such perspectives highlight that the spread of inaccurate or deceptive information is not merely a communication problem but also an epistemic challenge that may undermine knowledge systems and public trust.

Within contemporary research, scholars commonly distinguish between misinformation and disinformation based on the role of intentionality. Misinformation is

generally defined as false or misleading information that is shared without the intention to deceive (Fallis 2015). Such information may arise from misunderstanding, error, or the unintentional spread of inaccurate claims. In contrast, disinformation refers to false or misleading information that is deliberately fabricated or manipulated with the intention to deceive or mislead audiences (Fetzer 2004, Fallis 2015). Fetzer (2004) emphasises that disinformation involves the intentional use of false information to influence beliefs, while Fallis (2015) further clarifies that the defining characteristic of disinformation lies in its deceptive intent. Although this distinction is widely adopted, it is important to note that intentionality may be difficult to observe directly in empirical contexts. Nevertheless, the intentionality-based distinction remains a useful analytical framework for differentiating between unintentional and strategic forms of misleading communication.

Earlier research frequently used the term “fake news” to describe misleading information circulating in media environments. However, scholars have increasingly criticised this concept for its conceptual ambiguity. Gelfert (2018) argues that fake news refers to deliberately misleading information that imitates the format and appearance of legitimate journalism. Similarly, Tandoc et al. (2018) demonstrate that the term has been used inconsistently across studies, encompassing satire, parody, propaganda, fabricated news stories, and misleading journalistic practices. Because these forms of content differ substantially in terms of intent, structure, and social impact, the term “fake news” lacks analytical precision and has therefore been criticised as a problematic concept in academic research.

These conceptual distinctions also reflect an evolution in how problematic information has been understood in digital communication environments. Earlier discussions often relied on broad and loosely defined terms such as “fake news”, which encompassed a wide range of heterogeneous phenomena. Over time, scholars have increasingly adopted more precise distinctions between misinformation and disinformation in order to capture the role of intentionality in misleading communication. More recent work further expands this perspective by situating these categories within broader frameworks such as information disorder, which emphasise the interactions between actors, content, and technological infrastructures.

In this review, the distinction between misinformation and disinformation follows the intentionality-based framework widely adopted in the literature. Misinformation is treated as false or misleading information shared without the intention to deceive, whereas disinformation is understood as false or misleading information that is deliberately created or disseminated with the intention to mislead (Fallis 2015, Fetzer 2004). While the intention behind information dissemination may not always be directly observable in empirical settings, this distinction provides a useful conceptual foundation for analysing different forms of problematic information in socio-technical systems. However, because intentionality is often difficult to verify empirically, this review primarily adopts misinformation as the broader analytical category through which problematic information diffusion is examined, while disinformation remains conceptually important for distinguishing intentionally deceptive forms of communication.

While the distinction between misinformation and disinformation is conceptually grounded in intentionality, empirical analysis often relies on more observable forms of evaluation. In practice, particularly in case-based analyses, problematic information is commonly identified based on its inconsistency with established, verifiable, or widely accepted information rather than inferred intent. As a result, empirical studies frequently focus on the circulation, visibility, and diffusion dynamics of mis-

leading information regardless of whether intentional deception can be conclusively determined.

It is also important to note that the notion of “ground truth” may itself be context-dependent, particularly in politically or socially contested domains. As such, this review adopts a pragmatic approach, treating problematic information as content that contradicts widely accepted or verifiable claims, while recognising the limitations of such classifications.

However, while these conceptual and operational distinctions provide useful analytical clarity, they do not fully capture the diversity and complexity of problematic information in contemporary digital environments. In practice, misleading content often spans multiple categories and interacts across different forms, making it difficult to analyse solely within the boundaries of misinformation and disinformation.

To clarify this progression from conceptual distinction to a broader analytical perspective, Figure 2.1 illustrates how misinformation and disinformation are defined, operationalised, and subsequently extended toward system-level analysis.

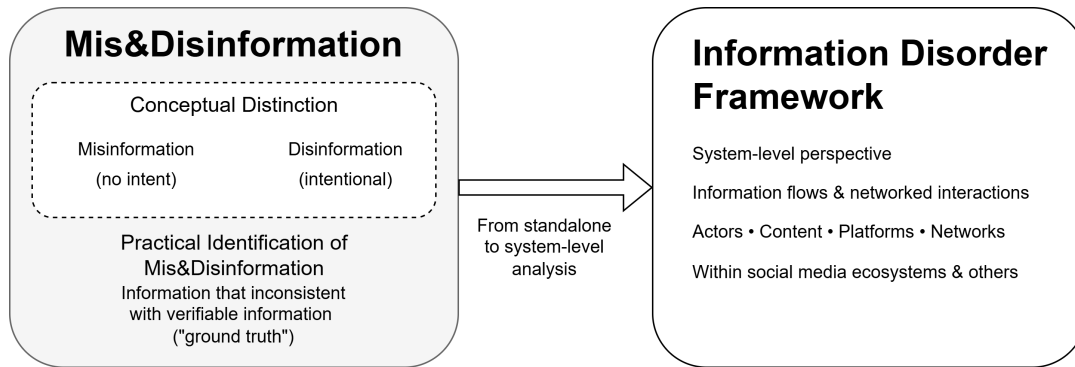


Figure 2.1: From standalone conceptual distinctions to system-level analysis of misinformation.

As shown in Figure 2.1, while misinformation and disinformation provide a useful starting point for analysing problematic information, their standalone application is insufficient for capturing the complexity of contemporary information environments.

Jack (2017) highlights, contemporary media environments encompass a wide range of related phenomena, including misinformation, disinformation, propaganda, rumours, and other forms of misleading narratives. This diversity suggests that problematic information cannot be adequately understood as isolated categories, but rather as part of a broader informational landscape.

Recognising this complexity, several scholars have proposed broader conceptual frameworks for understanding problematic information in digital environments. Building on this need for a more comprehensive perspective, Wardle & Derakhshan (2017) introduce the influential *information disorder* framework.¹ Rather than treating misleading information solely as isolated false content, such approaches emphasise the broader socio-technical environments within which information is produced, circulated, interpreted, and amplified.

Such a system-level perspective highlights that misleading information is not simply a property of individual pieces of content, but an emergent outcome of in-

¹The information disorder framework distinguishes between misinformation, disinformation, and malinformation while emphasising the relationships between actors, messages, and communication infrastructures within digital information environments. Related system-level dynamics concerning information circulation, network interaction, and feedback processes are further discussed in Sections 2.2 and 2.3.

teractions between information producers, technological platforms, and audiences. Understanding these interactions provides an essential foundation for analysing how misinformation spreads within contemporary communication networks. The following section therefore introduces the information disorder framework in greater detail and examines how actors, information flows, and technological infrastructures interact within modern information environments.

2.2 Misinformation as a System

Building on the information disorder framework, misinformation and disinformation can be further understood as phenomena embedded in socio-technical communication systems composed of actors, information objects, and network structures (Wardle & Derakhshan 2017). Rather than emerging in isolation, problematic information circulates through environments in which different participants produce, interpret, amplify, and contest narratives. Analysing these elements is therefore essential for understanding how misleading information acquires visibility, credibility, and influence within contemporary digital communication systems (Wardle & Derakhshan 2017, Frischlich et al. 2025).

To provide a structured overview of these core components, Table 2.1 summarises the key elements of misinformation as a socio-technical system.

Table 2.1: Core components of misinformation as a socio-technical system

Component	Description	Examples
Actors	Participants involved in producing, sharing, and interpreting information	Users, media organisations, political actors, bots
Information objects	Content carrying narratives and meaning	News, rumours, memes, images, videos
Network structures	Patterns of connections shaping information diffusion	Follower networks, communities, clusters
Information flows	Processes through which information circulates and spreads	Reposting, recommendation, sharing chains

As shown in Table 2.1, these components jointly constitute the structural basis of misinformation systems, which are elaborated in the following subsections.

Actors in digital information environments. A wide range of actors participate in the production and circulation of problematic information. These include individual users, media organisations, political actors, activist groups, coordinated influence networks, and automated accounts such as bots. In digital platforms, users are not merely passive audiences but also active distributors of information through sharing, reposting, commenting, and endorsing content.² At the same time, institutional and political actors may strategically shape narratives in pursuit of ideological, political, or economic goals. Empirical studies further suggest that coordinated groups can exploit crises and contentious events to increase the visibility

²The active role of users in amplifying and circulating misinformation became particularly visible during the COVID-19 pandemic, where large-scale sharing behaviour significantly contributed to the rapid diffusion of misleading health information across digital platforms. Related psychological and behavioural mechanisms underlying such participation are further discussed in Chapter 5.

of particular narratives (Bailo et al. 2024). Thus, misinformation ecosystems should be understood as environments structured by heterogeneous actors with different capacities, intentions, and communicative roles.

Information objects and narrative structures. Within these systems, misinformation are carried by diverse information objects, including news stories, rumours, memes, images, videos, and short social media posts. These objects are not simply neutral containers of content; rather, they often embed narrative and interpretive structures that shape how events are framed and understood. Research on misinformation ³ suggests that such content often gains traction when it resonates with pre-existing beliefs, emotional responses, or ideological commitments (Gelfert 2018, Lewandowsky et al. 2017). In this sense, misinformation should be analysed not only as false content but also as a narrative form that interacts with the cognitive and social contexts of audiences.

Network structures of online communication. The circulation of information is shaped by the underlying network structures of digital communication. Users are connected through follower relations, friendship ties, reply networks, and community clusters, and these connections influence how far and how quickly information travels. Research in network science shows that highly connected hubs, clustered communities, and dense interaction patterns play important roles in shaping information visibility and diffusion (Kumar & Sinha 2021). Bakshy, Messing, and Adamic (Bakshy et al. 2015) further demonstrate that patterns of exposure to information in online platforms are affected by network ties and ideological selectivity, while Del Vicario et al. (2016) show, through analyses of Facebook communities surrounding scientific and conspiracy-related content, that misinformation often circulates within highly polarised and internally reinforcing communities. These findings suggest that the networked organisation of communication is not simply a background condition but a constitutive part of how misinformation spreads.

Information flows and networked circulation. When actors interact through networked platforms, information flows emerge through reposting, recommendation, commenting, and imitation. Such flows create pathways through which narratives move across communities and sometimes across platforms. Early research on information contagion demonstrated that online social networks shape not only the speed but also the scale and form of information cascades (Lerman & Ghosh 2010). More recent empirical work has shown that misleading information may spread farther, faster, and more broadly than accurate information under certain conditions, particularly when amplified by network structures and audience engagement (Vosoughi et al. 2018a). Taken together, these studies indicate that misinformation should be understood as a networked process rather than a property of isolated messages.

Understanding the relationships between actors, information objects, and network structures therefore provides a necessary foundation for analysing how misinformation circulates in contemporary digital environments. The following section extends this system perspective by examining how feedback mechanisms further amplify, stabilise, or transform the circulation of problematic information within online communication systems.

³Related research may also use terms such as "fake news", "misleading information", "false information", or "disinformation" depending on disciplinary focus, empirical context, and conceptual definitions. In this review, misinformation is adopted as the broader analytical category unless distinctions regarding intentionality are specifically relevant.

2.3 Feedback Loops in Information Systems

While actors, information, and networks describe the structural components of on-line information systems, the spread of misinformation is strongly shaped by feedback mechanisms that continuously reinforce information circulation. From a cybernetic perspective, communication environments can be understood as systems in which outputs influence future inputs, producing recursive processes that shape the visibility and diffusion of information over time. In digital media environments, these feedback dynamics emerge through the interactions between user behaviour, algorithmic curation⁴, and network structures.

To illustrate how these feedback mechanisms operate as a unified system, Figure 2.2 presents a system-level model of recursive interactions between users, platform algorithms, and communities.

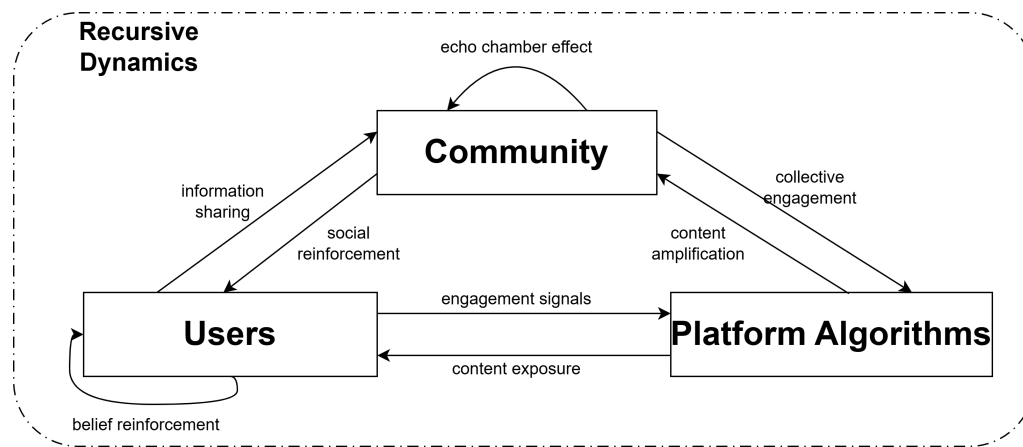


Figure 2.2: Recursive dynamics of misinformation, showing feedback loops in systems between users, platform algorithms, and communities.

As shown in Figure 2.2, misinformation diffusion emerges from multiple interacting feedback loops. User engagement generates behavioural signals that influence algorithmic ranking, while algorithmic exposure shapes the visibility of content and subsequent user interactions. At the same time, community-level dynamics reinforce shared beliefs through repeated exposure and social validation, further stabilising and amplifying information within networked environments. Together, these interactions form a self-reinforcing system in which engagement, visibility, and community processes mutually amplify one another. These interconnected feedback processes can be analytically distinguished into several key mechanisms, which are discussed in turn below.

Engagement-driven feedback. One important feedback mechanism arises from user engagement behaviour. Digital platforms frequently use engagement metrics such as likes, shares, comments, and reposts to evaluate the relevance or popularity of information. Content that generates stronger reactions tends to receive greater visibility in platform ranking systems. Empirical research has shown that misinformation often produces higher levels of novelty and emotional response, which

⁴Algorithmic curation refers to the automated selection, ranking, and prioritisation of content by digital platforms based on behavioural signals, engagement metrics, and platform-specific recommendation systems.

can increase user engagement and accelerate information diffusion (Vosoughi et al. 2018a). Cognitive research further suggests that individuals are more likely to interact with information that aligns with existing beliefs or provokes strong emotional reactions (Lewandowsky et al. 2017). Because engagement signals⁵ are repeatedly incorporated into platform ranking algorithms, highly engaging information may experience self-reinforcing visibility, generating feedback loops that amplify the circulation of misinformation across social networks.

Algorithmic amplification. A second feedback mechanism emerges through algorithmic content curation. Social media platforms rely on ranking algorithms and recommendation systems to organise large volumes of information and personalise user feeds. These algorithms typically prioritise content that is predicted to maximise user engagement and platform activity (Gillespie 2014). As a result, information that initially receives high engagement may be promoted to wider audiences, increasing its exposure and generating additional engagement signals. Studies of online information exposure demonstrate that algorithmic filtering⁶ plays a major role in shaping the information environments that users encounter (Bakshy et al. 2015). From a systems perspective, algorithmic ranking⁷ mechanisms therefore function as key feedback structures that influence which information becomes widely visible within digital communication environments. In the context of misinformation, such mechanisms may unintentionally amplify misleading or sensational content because of its strong engagement signals (Shu et al. 2017a).

Community reinforcement and echo chambers. Network structures further contribute to feedback dynamics in online information systems. Users in social networks frequently interact with others who share similar beliefs, interests, or ideological orientations, a process often described as homophily. Such interaction patterns can lead to the formation of echo chambers, where information circulates predominantly within like-minded communities. Empirical analyses of social media networks show that users tend to cluster into communities that repeatedly share similar narratives and information sources (Del Vicario et al. 2016). Within these communities, repeated exposure to similar viewpoints can reinforce existing beliefs and strengthen group-level interpretations of events. Research on online political communication further suggests that ideological clustering may influence the types of information individuals encounter and share within social networks (Barberá et al. 2015). These community-level dynamics can therefore contribute to the persistence and reinforcement of misinformation narratives across online environments (Cinelli et al. 2021).

Recursive dynamics of misinformation systems. Taken together, these mechanisms illustrate how misinformation diffusion emerges from recursive interactions between user behaviour, algorithmic systems, and network structures. Rather than spreading in a purely linear manner, misinformation often propagates through complex socio-technical feedback loops in which engagement signals, algorithmic visibility, and community interactions mutually reinforce one another. From a systems perspective, online communication environments can therefore be conceptualised as cybernetic information systems in which feedback processes continuously reshape

⁵Engagement signals refer to measurable forms of user interaction with content, such as likes, shares, comments, watch time, clicks, and reposting behaviour, which platforms use to evaluate and prioritise visibility within recommendation systems.

⁶Algorithmic filtering refers to the automated selection or exclusion of content from a larger pool of available information based on platform rules, behavioural data, or recommendation criteria.

⁷Algorithmic ranking refers to the ordering and prioritisation of selected content according to predicted relevance, engagement potential, or other platform-specific objectives.

the circulation of information (Kumar & Sinha 2021). Recent reviews of misinformation research further emphasise that understanding these systemic dynamics is essential for analysing how misleading information evolves and spreads across digital platforms (Xu et al. 2025).

2.4 What is Misinformation?

Returning to the question that opened this chapter—What is misinformation?—the discussions above suggest that no single definition is sufficient to fully capture the complexity of misinformation within contemporary communication environments. While misinformation is often treated as a property of individual pieces of information, the perspectives reviewed in this chapter suggest that understanding misinformation requires attention not only to information itself but also to the systems within which it exists.

From this perspective, misinformation is understood not simply as a fixed informational object but as a dynamic phenomenon shaped by interactions between information, actors, communication structures, and feedback processes operating within broader socio-technical systems. This understanding provides the conceptual foundation for the remainder of this thesis and serves as the starting point for the chapters that follow.

Chapter 3

Understanding Misinformation Through Diffusion

Chapter 2 argued that misinformation cannot be understood solely as a property of individual pieces of information, but must also be understood in relation to the socio-technical systems within which it exists. If misinformation emerges through interactions between information and communication systems, an important next question follows: how does information move through these systems?

To address this question, this chapter examines how information diffusion has been understood within existing literature. Through both theoretical diffusion models and empirical studies, the chapter examines how information spreads through communication systems and whether misinformation displays distinctive patterns of diffusion compared with other forms of information.

3.1 Foundational Models of Information Diffusion

Understanding how information spreads within social networks has become a central concern in network science and computational social science ([Lazer et al. 2018](#), [Kumar & Sinha 2021](#)). If misinformation is understood as a phenomenon emerging through interactions between information and communication systems, then understanding how information moves within these systems becomes an important part of understanding misinformation itself.

One way researchers have approached this question is through diffusion models. Rather than focusing on the content of information alone, diffusion models provide structured ways of examining the relationships between information, individuals, and networked communication environments. Different models emphasise different mechanisms of diffusion, including transmission, social influence, and network structure, offering distinct perspectives on how information propagates through communication systems.

The following sections examine three major traditions of diffusion modelling—epidemic-based, threshold and cascade, and network-based approaches—and discuss how each contributes to understanding information diffusion within communication systems.

3.1.1 Epidemic-Based Models

Epidemic-based models¹ represent one of the earliest and most influential paradigms for modelling information diffusion. At a conceptual level, these models treat information diffusion as a process in which individuals “pass on” information upon contact, similar to how a contagious disease spreads in a population. In this sense, individuals who encounter information may adopt and share it, and in some cases eventually stop participating in the diffusion process.

Drawing on classical epidemiology, models such as the Susceptible–Infected (SI), Susceptible–Infected–Susceptible (SIS), and Susceptible–Infected–Recovered (SIR) frameworks conceptualise information as analogous to a contagious disease that spreads through contact between individuals ([Kermack & McKendrick 1927](#), [Pastor-Satorras et al. 2015](#)). These models differ primarily in how they represent what happens after an individual adopts information: SI assumes persistent spreading once adopted, SIS allows individuals to repeatedly adopt and share information, and SIR introduces a recovery state in which individuals eventually stop participating in the diffusion process.

Within this framework, diffusion is typically modelled as a state transition process, where individuals move between discrete states (e.g., susceptible, infected, recovered) based on transmission probabilities and network structure. This perspective enables researchers to analyse diffusion dynamics in terms of infection rates, thresholds, and equilibrium states, and has been widely applied to the study of large-scale information cascades and viral spreading phenomena in networks ([Pastor-Satorras et al. 2015](#)).

Despite their mathematical tractability, epidemic-based models rely on assumptions that limit their ability to capture the complexity of real-world information environments. In particular, they treat individuals as passive carriers of information and assume that exposure alone is sufficient to trigger transmission. Such assumptions overlook the role of cognitive processes, selective attention, and individual decision-making, which are central to how users engage with information in digital contexts. Furthermore, these models typically consider the spread of a single piece of information in isolation, thereby neglecting the presence of competing narratives and the broader informational ecosystem.

3.1.2 Threshold and Cascade Models

To address the limitations of epidemic-based approaches, threshold and cascade models introduce a more behaviourally grounded perspective on information diffusion ([Kumar & Sinha 2021](#)). In contrast to passive transmission, these models assume that individuals actively decide whether to adopt or share information based on the behaviour of others in their social network. In intuitive terms, this means that people do not immediately adopt information upon exposure, but instead wait until enough of their peers have already done so.²

Threshold models, most notably proposed by [Granovetter \(1978\)](#), conceptualise diffusion as a process driven by social influence and individual decision-making. In this framework, each individual is associated with a threshold that represents the

¹Epidemic-based models are mathematical models originally developed to study the spread of infectious diseases, later adapted to analyse how information propagates through social networks.

²For example, a user may initially ignore a rumour encountered online, but become more likely to believe or share it after seeing many friends, influencers, or community members repeatedly endorse the same claim.

proportion of their peers who must adopt a behaviour before they do so themselves. Diffusion occurs when the cumulative influence from neighbouring nodes exceeds this threshold, leading to adoption. This approach captures the role of social reinforcement and highlights that individuals are not passive recipients of information, but instead respond to collective behavioural signals within their local network.

While threshold models focus on individual decision rules, cascade models emphasise the collective outcomes of these local decisions. In particular, cascade models examine how small initial events can trigger large-scale diffusion, helping to explain why some pieces of information suddenly become widespread or “go viral”. The model developed by [Watts \(2002\)](#) demonstrates that diffusion is highly sensitive to network structure, and that global spreading events can emerge from relatively minor perturbations under certain conditions.

Despite incorporating elements of social influence and decision-making, threshold and cascade models still rely on simplifying assumptions that limit their applicability to contemporary information environments. For example, thresholds are often treated as static and homogeneous, and decision processes are typically modelled in a highly simplified manner. Moreover, these models generally do not account for platform-mediated exposure³, algorithmic curation, or the presence of multiple competing information streams.

3.1.3 Network-Based Diffusion Models

Building on epidemic and threshold-based approaches, network-based diffusion models provide a more formalised framework for analysing how information propagates through complex network structures ([Kumar & Sinha 2021](#)). In simple terms, these models focus on how the pattern of connections between individuals shapes how far and how quickly information spreads. Rather than only considering individual behaviour, they emphasise that diffusion outcomes depend heavily on network structure.

Two of the most widely studied models in this domain are the Independent Cascade (IC) model and the Linear Threshold (LT) model, both introduced in the context of influence maximisation by [Kempe et al. \(2003\)](#). In the IC model, an activated node has a single opportunity to influence each of its neighbours with a given probability, reflecting a stochastic and independent transmission process. In contrast, the LT model assumes that each node adopts a behaviour when the cumulative influence from its neighbours exceeds a predefined threshold, thereby integrating elements of social reinforcement into the diffusion process.

A key strength of network-based diffusion models lies in their ability to capture how network structure shapes information spread. Properties such as node centrality, clustering, and connectivity patterns play a crucial role in determining diffusion outcomes. These models have therefore been widely applied to identify influential nodes and analyse large-scale propagation processes in online social networks ([Kitsak et al. 2010](#)).

However, despite their increased structural sophistication, network-based models still rely on simplifying assumptions that limit their applicability to real-world

³Platform-mediated exposure refers to situations in which users encounter information through platform-controlled systems such as recommendation algorithms, ranking mechanisms, and curated feeds, rather than through direct or neutral information access. Related platform dynamics concerning visibility, recommendation systems, and algorithmic amplification are further discussed in Chapter 6.

information environments. In particular, they often assume static network structures, fixed transmission probabilities, and independent diffusion processes, thereby overlooking the dynamic and adaptive nature of online communication systems. Moreover, these models typically do not account for platform-mediated exposure, algorithmic recommendation systems, or the presence of multiple competing information streams.

3.1.4 Summary and Limitations of Foundational Diffusion Models

To clarify how these different models capture distinct aspects of diffusion, Table 3.1 provides a comparative overview of their underlying assumptions, strengths, and limitations.

Although epidemic, threshold, and network-based models offer progressively richer accounts of information diffusion, they rely on simplifying assumptions that limit their explanatory power in contemporary socio-technical environments. In particular, they often overlook cognitive processes, platform-mediated exposure, feedback mechanisms, and the adaptive nature of communication systems. As a result, they provide only a partial understanding of how information, especially misinformation, spreads in real-world environments. Nevertheless, these models establish important foundations for examining diffusion processes and identifying diffusion characteristics through which misinformation may become observable within communication systems.

Table 3.1: Comparison of foundational diffusion models

Model Type	Analytical Perspective	Core Assumptions	Strengths	Limitations	Typical Use
Epidemic-based (SI, SIS, SIR)	Transmission	Passive spread; exposure leads to transmission; homogeneous interactions	Clear mathematical foundation; captures large-scale spreading dynamics	Ignores cognition, decision-making, and competing information	Large-scale diffusion modelling
Threshold models	Behavioural	Adoption depends on social influence exceeding thresholds	Captures social reinforcement and decision processes	Static assumptions; simplified behaviour	Peer influence analysis
Cascade models	Behavioural-structural	Small shocks can trigger large cascades depending on structure	Explains viral diffusion and tipping dynamics	Simplified behaviour; limited realism	Viral spread and cascade analysis
Network-based (IC, LT)	Structural	Diffusion follows network topology with probabilistic activation	Captures structural influence and propagation pathways	Assumes static networks; ignores platform dynamics	Identifying influential nodes

As shown in Table 3.1, these models can be understood as complementary analytical perspectives, each capturing different mechanisms of diffusion, from passive transmission to behavioural decision-making and structural influence.

3.2 Influence Maximization in Social Networks

Building on the diffusion models discussed in the previous section, it is useful to consider not only how information spreads, but also how its spread can be influenced. In many real-world situations, such as marketing campaigns, public information

initiatives, or online communication, the goal is often to maximise the reach of information rather than simply observe its diffusion. This leads to the problem of influence maximization (IM), which seeks to identify a small number of influential individuals whose participation can trigger the widest possible spread of information throughout a network.

Formally introduced by [Kempe et al. \(2003\)](#), the problem is defined as selecting a seed set of size k (i.e., a limited number of initial users from which diffusion begins) that maximises the expected number of activated nodes (i.e., the average number of individuals reached under probabilistic diffusion).

Figure 3.1 illustrates this process at a conceptual level. Starting from an initial set of seed nodes, information propagates through the network according to a given diffusion model (e.g., Independent Cascade or Linear Threshold), resulting in a set of activated nodes. The influence of a seed set is evaluated based on the expected spread of activation, and the selection of seed nodes is iteratively refined using greedy optimisation⁴ strategies.

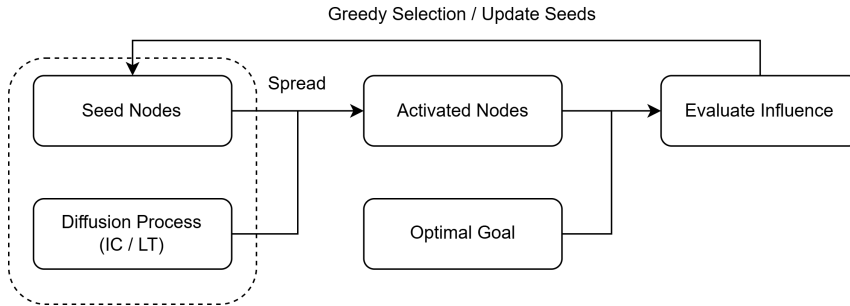


Figure 3.1: Conceptual illustration of influence maximization. A set of seed nodes initiates diffusion through the network under a specified diffusion model (e.g., IC or LT). The resulting spread is evaluated, and seed selection is iteratively refined to maximise the expected number of activated nodes.

This formulation has become the foundational framework for studying information propagation at scale. The IM problem is inherently grounded in the diffusion processes discussed in Section 3.1. In other words, determining which individuals are “influential” depends on how information is assumed to spread through the network. The definition of influence spread therefore depends directly on diffusion models, typically operationalised using probabilistic frameworks such as the Independent Cascade (IC) and Linear Threshold (LT) models. As discussed in [Li et al. \(2018\)](#), these models not only describe propagation dynamics but also define the optimisation objective of influence maximization, thereby tightly coupling modelling assumptions with algorithmic design.

Despite its intuitive formulation, influence maximization is computationally intractable. This means that it is extremely difficult to identify the optimal set of initial nodes, particularly in large-scale networks. Moreover, even estimating how far information will spread from a given set of nodes requires considering a vast number of possible diffusion scenarios, as each potential interaction can lead to different propagation outcomes ([Kempe et al. 2003](#), [Li et al. 2018](#)).

To address these challenges, research has focused on approximation algorithms that balance scalability and solution quality. A key theoretical insight is that, under

⁴Greedy optimisation refers to an iterative strategy that repeatedly selects the locally most beneficial option at each step in order to progressively improve the overall solution.

widely used diffusion models, the influence function is both monotone (i.e., adding more seed nodes will not decrease the total spread) and submodular (i.e., exhibiting diminishing marginal returns). In intuitive terms, this implies that selecting the most impactful individuals first yields the greatest benefit, while adding additional nodes provides progressively smaller gains. These properties enable the use of greedy algorithms with provable approximation guarantees (Kempe et al. 2003), establishing the greedy framework as a cornerstone of influence maximization research.

However, structural heuristics such as degree centrality or PageRank do not necessarily align with actual diffusion outcomes. As argued by Borgatti (2005), centrality measures capture specific notions of flow within networks but may fail to represent influence under stochastic propagation processes. Empirical evidence further challenges simplistic assumptions about influence. For example, Kitsak et al. (2010) demonstrate that nodes located in the network core can be more influential than high-degree hubs, while large-scale experiments by Bakshy et al. (2012) show that information cascades are highly heterogeneous and difficult to predict.

Taken together, these findings suggest that influence is shaped by more than network structure alone. How information spreads also depends on diffusion dynamics and the behaviour of the people involved. At the same time, most classical influence maximization models assume that only a single piece of information is spreading through a network. In reality, however, communication environments are often characterised by multiple competing narratives. This limitation has motivated the development of competitive diffusion models, which are discussed in the following section.

3.3 Competitive Information Diffusion and Multi-Agent Dynamics

While classical information diffusion models typically assume a single source of influence propagating through a network, real-world information environments are inherently competitive. Multiple actors, such as individuals, organisations, media outlets, or platforms, often attempt to influence the same population simultaneously, sometimes with conflicting objectives. As a result, users are frequently exposed to competing information streams and must decide which information to attend to, trust, or adopt (Solanki et al. 2025). In this context, diffusion becomes a competition for attention, credibility, and influence rather than a simple spreading process, transforming information diffusion into a strategic interaction problem.

This competitive setting is formalized in the framework of *Competitive Influence Maximization (CIM)*, where multiple sources aim to maximize their own influence spread while limiting that of others (Bharathi et al. 2007, Kempe et al. 2003). Unlike single-source diffusion, where activation is typically monotonic and independent, competitive diffusion introduces exclusivity—nodes generally adopt only one competing influence. As a result, diffusion outcomes become interdependent, and the effectiveness of one agent’s strategy is conditioned on the presence and behavior of others. In simple terms, this means that the success of one piece of information depends not only on how it spreads, but also on how it competes with alternative information.

Figure 3.2 illustrates this mechanism at the node level. A single user receives competing signals from multiple information sources and must make an adoption decision based on their relative influence. For instance, a user may encounter both

a misleading claim and a corrective explanation, and their final choice depends on which influence is stronger or more persuasive. Crucially, once a node adopts one influence, it typically does not adopt competing alternatives, reflecting the exclusivity characteristic of competitive diffusion.

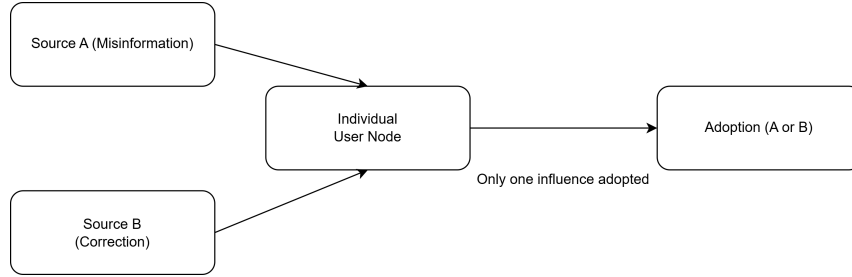


Figure 3.2: Node-level illustration of competitive diffusion. A single user receives competing information from multiple sources and makes an adoption decision based on relative influence. Only one competing influence is adopted.

To better understand how such competitive interactions unfold, researchers have proposed several extensions of classical diffusion models, including the Competitive Independent Cascade (CIC) and Competitive Linear Threshold (CLT) models. These models explicitly incorporate multiple competing cascades, allowing nodes to evaluate incoming influence from different sources and adopt based on thresholds, probabilities, or dominance rules. For instance, competitive threshold-based models enable nodes to compare accumulated influence from competing sources and select the dominant one based on relative influence strength (Pham et al. 2019).

From an intuitive perspective, competitive diffusion models describe how individuals weigh competing signals before making adoption decisions. At the network level, these local decisions collectively produce large-scale diffusion patterns in which different regions may become dominated by different information sources. Figure 3.3 illustrates this process, showing how interactions among competing influences generate feedback-driven dynamics and diverse diffusion outcomes across the network. Even when users experience similar forms of exposure, variations in local interactions and diffusion pathways can lead to substantially different network-level configurations.

Beyond propagation mechanisms, network structure plays a critical role in shaping competitive diffusion outcomes. Identifying influential nodes becomes more complex in competitive settings, as influence is inherently relational rather than absolute. Nodes that are highly central in isolation may lose effectiveness if competing agents occupy structurally overlapping regions. Empirical and theoretical studies suggest that nodes located at structural bridges or community boundaries can significantly alter competitive dynamics by redirecting influence flows across the network (Kitsak et al. 2010, Borgatti 2005, Bakshy et al. 2012). In simple terms, this means that where a node is located in the network can matter just as much as how influential it is.

From a computational perspective, competitive diffusion introduces substantial challenges. Influence maximization is already NP-hard⁵ in single-source settings⁶ (Kempe et al. 2003), and the introduction of multiple competing agents further

⁵NP-hard refers to a class of computational problems for which no known algorithm can efficiently guarantee an optimal solution as problem size grows.

⁶Single-source settings refer to diffusion environments in which information originates from only one source or competing actor rather than multiple simultaneously interacting sources.

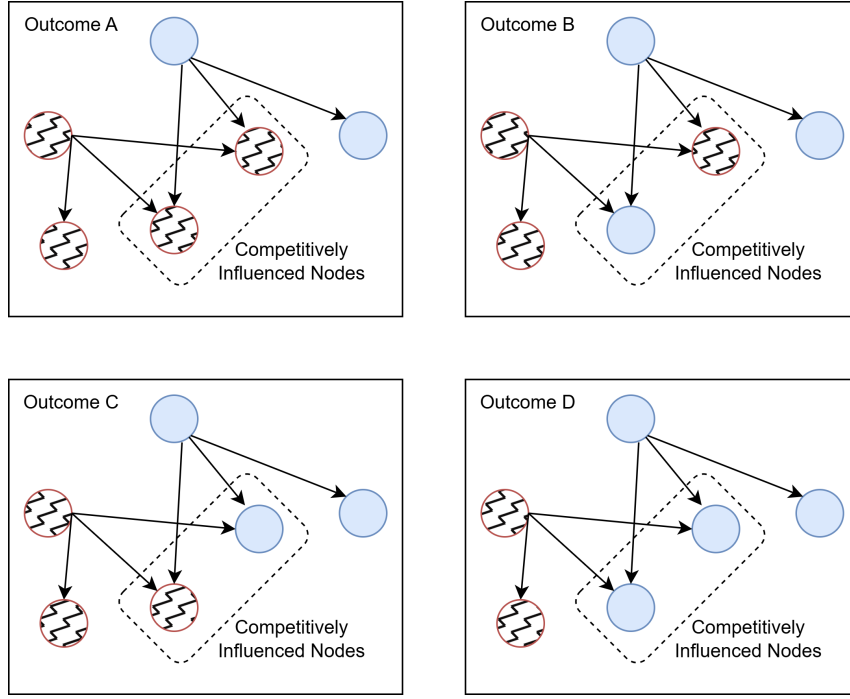


Figure 3.3: Illustration of competitive diffusion outcomes in competitive influence maximization (CIM). Nodes exposed to competing information sources may adopt different influences depending on local network interactions and diffusion pathways. Although competitively influenced nodes experience similar exposure conditions, different diffusion outcomes may emerge across the network.

complicates the problem. In particular, competitive influence functions are often neither monotonic nor submodular, which undermines the effectiveness of standard greedy approximation algorithms. In intuitive terms, this means that finding the best set of initial nodes becomes much harder when multiple competing influences are present, as outcomes depend not only on one’s own strategy but also on the actions of competitors. As highlighted in recent surveys, scalability, uncertainty, and dynamic network conditions remain key challenges in competitive influence maximization (Solanki et al. 2025).

Beyond competitive settings, recent research has extended influence maximization into several important directions, including context-aware influence maximization (CAIM), trust-based models, dynamic network settings, and learning-based approaches. These extensions reflect the recognition that diffusion processes are shaped by contextual factors, temporal evolution, and adaptive behaviors rather than static structures alone. For example, dynamic influence maximization considers evolving network structures where connections and influence pathways change over time (Zhuang et al. 2013). Trust-based models account for the credibility and reliability of information sources, while context-aware approaches incorporate user attributes and environmental factors into diffusion processes. In addition, recent advances in machine learning have led to learning-based influence maximization methods, where deep graph representation learning techniques are used to capture complex structural and contextual patterns in networks (Ling et al. 2023). While not explicitly framed as context-aware influence maximization, such approaches are capable of implicitly incorporating contextual information through representation learning.

Furthermore, recent work has begun to integrate competitive and targeted per-

spectives, where influence strategies are designed not only to maximize spread but also to focus on specific groups or outperform competing sources (Liang et al. 2023). However, despite these advances, most of these approaches still operate under a single-agent assumption. In contrast, competitive influence maximization explicitly models interactions among multiple agents, making it more suitable for capturing real-world information ecosystems.

From a broader perspective, competitive diffusion can be understood as a *multi-agent cybernetic system*, in which multiple actors continuously adjust their strategies in response to others within a feedback-driven environment. As illustrated in Figures 3.2 and 3.3, competitive diffusion operates through recursive interactions between local decision-making and global network dynamics. In simple terms, this means that individual choices and overall network patterns continuously influence each other over time.

In the context of misinformation, these dynamics become particularly significant, as truthful and false information compete for attention, credibility, and adoption within shared communication networks. Diffusion therefore becomes more than a simple spreading process, involving strategic intervention, counter-messaging, and ongoing interactions between competing actors. However, although competitive diffusion models extend earlier frameworks by incorporating strategic interaction and multi-agent dynamics, they remain simplified representations of highly complex communication environments. Real-world misinformation diffusion is also shaped by heterogeneous user behaviour, temporal variation, contextual amplification, and platform-mediated visibility, factors that are difficult to fully capture through diffusion models alone.

3.4 Diffusion Complexity and Real-World Dynamics

The diffusion models discussed in previous sections provide useful frameworks for understanding how information spreads and how influence operates within networked environments. However, real-world misinformation diffusion is often more dynamic, diverse, and context-dependent than abstract diffusion models can fully capture. Table 3.2 compares the key assumptions underlying different diffusion frameworks and highlights how successive models incorporate additional dimensions such as strategic intervention, competing information, and explicit decision-making processes. Despite these advances, important characteristics of contemporary communication systems—including platform mediation, behavioural variability, contextual amplification, and evolving information environments—remain only partially represented.

These limitations become particularly visible when examining empirical diffusion patterns observed in real-world misinformation environments. The following subsections therefore explore large-scale propagation dynamics, cascade structures, temporal diffusion behaviour, and contextual amplification processes across digital communication systems.

⁷Refers to whether diffusion models incorporate cognitive, emotional, or interpretive processes influencing user adoption and sharing behaviour.

⁸Refers to whether models account for platform-controlled exposure mechanisms such as recommendation systems, ranking algorithms, or algorithmic curation.

Table 3.2: Comparison of diffusion models across key modelling assumptions

Assumption / Feature	Basic Diffusion	Influence Maximization	Competitive Influence Maximization
Network structure considered	✓	✓	✓
Dynamic network structure	×	×	×
Single information source	✓	✓	×
Competing information	×	×	✓
Strategic intervention (seed selection)	×	✓	✓
Node decision-making	limited	partial	explicit
User cognition considered ⁷	×	×	limited
Platform mediation ⁸	×	×	×

3.4.1 Large-Scale Diffusion Patterns

Large-scale observational studies provide critical empirical evidence for understanding how misinformation spreads in real-world information environments. One of the most influential studies by Vosoughi et al. (2018a), based on Twitter data collected between 2006 and 2017, analysed approximately 126,000 information cascades involving around 3 million users. By reconstructing diffusion trees and comparing propagation characteristics, the study finds that false information spreads significantly faster, reaches a wider audience, and penetrates deeper into networks than true information.

Subsequent research provides further support for these patterns across different platforms. For example, Del Vicario et al. (2016), using Facebook data, analyse user interactions with scientific and conspiracy-related content and show that misinformation tends to spread within homophilous communities⁹, where users preferentially engage with like-minded information. This reinforces internal diffusion and facilitates the persistence of misleading narratives. Similarly, Shao et al. (2018), based on Twitter data, identify the role of automated accounts (social bots) in the early stages of diffusion, demonstrating that bots disproportionately amplify low-credibility content by increasing its visibility and accelerating its initial spread before human users further propagate it.

These empirical findings suggest that misinformation diffusion is not purely a passive process but is shaped by behavioural, structural, and technological factors. In particular, the higher diffusion rates of misinformation are often attributed to its novelty and emotional appeal, which increase user engagement and sharing behaviour.

However, not all studies support a uniform pattern of large-scale diffusion. Research on information cascades by Goel et al. (2016), based on large-scale online diffusion data, shows that the vast majority of content does not spread widely, with only a small fraction achieving large-scale virality. This suggests that extreme diffusion events are relatively rare and that observed large cascades may represent exceptional rather than typical cases. In addition, Bakshy et al. (2015), analysing

⁹Homophilous communities refer to social groups in which individuals with similar beliefs, identities, or behavioural characteristics are more likely to connect and interact with one another.

Facebook news feed data from over 10 million users, find that exposure to ideologically diverse information still occurs, indicating that diffusion patterns are shaped by both algorithmic curation and user choice rather than being confined to fully isolated echo chambers.

Taken together, these findings indicate that while misinformation can exhibit strong diffusion characteristics, its spread is neither uniform nor universal. Instead, diffusion outcomes emerge from the interaction between content properties, user behaviour, network structure, and platform-specific dynamics. This complexity challenges simplified assumptions in classical diffusion models and highlights the need to analyse information spread as a context-dependent and system-level phenomenon.

3.4.2 Structural Characteristics of Information Cascades

While large-scale studies establish differences in the overall reach and speed of misinformation diffusion, understanding how information propagates requires examining the structural patterns of cascades. Rather than spreading uniformly, information diffusion typically exhibits structured pathways characterised by depth, branching, and clustering.

Consistent with prior findings ([Vosoughi et al. 2018a](#)), misinformation tends to form deeper and more complex cascades, indicating that its propagation often occurs through multi-step processes rather than shallow, broadcast-like dissemination. This observation is closely related to the concept of structural virality, which characterises diffusion based on whether information spreads through broad one-step exposure or through successive layers of re-sharing ([Goel et al. 2016](#)).

However, the emergence of such cascade structures can be explained through multiple mechanisms that operate simultaneously at different levels of the network. While these mechanisms can be examined individually, their combined effects are not easily captured through linear description alone. To provide a clearer and more integrated view, Figure 3.4 illustrates how these mechanisms interact to produce the observed cascade structures.

From a diffusion mechanism perspective, information spread is not always triggered by a single exposure. In their analysis of Twitter hashtag diffusion, [Romero et al. \(2011\)](#) examine how the probability of adoption changes with the number of prior exposures from social contacts. Their results show that, for many types of information, adoption likelihood increases with repeated exposures, indicating a process of social reinforcement. As illustrated in Figure 3.4, this mechanism manifests as multi-step propagation, where information spreads through successive layers of interaction.

Another explanation highlights the role of network topology, suggesting that cascade formation is shaped by the structural positions of nodes. Using k-core decomposition to simulate spreading processes, [Kitsak et al. \(2010\)](#) compare the influence of nodes based on degree and structural centrality. Their findings show that nodes located in the network core are more effective at spreading information than highly connected hubs, as they are better positioned to reach diverse parts of the network. In Figure 3.4, this is reflected in cross-network reach.

At the community level, cascade structures are further influenced by homophily and echo chamber dynamics. Through cross-platform analysis of user interactions on Twitter, Facebook, Reddit, and YouTube, [Cinelli et al. \(2021\)](#) measure the extent to which users are exposed to like-minded content and quantify the degree of community polarisation. Their results show that users tend to interact within ideologically

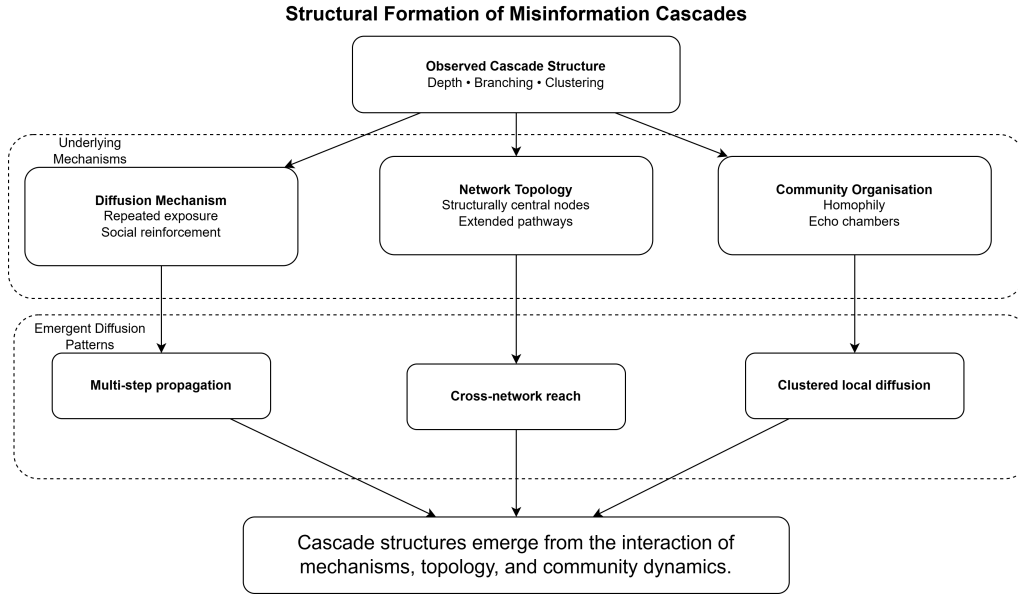


Figure 3.4: Conceptual illustration of the structural formation of misinformation cascades. Observed cascade structures emerge from the interaction between diffusion mechanisms, network topology, and community organisation, each producing distinct diffusion patterns that jointly shape overall propagation.

similar groups, leading to clustered diffusion patterns where information circulates repeatedly within communities but spreads less effectively across them (Del Vicario et al. 2016). As shown in Figure 3.4, this produces clustered local diffusion.

Overall, these findings suggest that cascade structures are not the result of a single process, but emerge from the interaction between diffusion mechanisms, network topology, and community organisation. As summarised in Figure 3.4, these mechanisms generate distinct intermediate patterns—multi-step propagation, cross-network reach, and clustered diffusion—which collectively give rise to the observed cascade structures. This highlights that information diffusion is not simply a linear or centralised process, but an emergent property of complex socio-technical systems.

3.4.3 Temporal Dynamics of Misinformation

Building on the structural patterns discussed in the previous section, it is equally important to examine how information diffusion unfolds over time. Rather than occurring at a constant or gradual rate, misinformation diffusion is typically characterised by rapid early growth followed by uneven patterns of propagation and decline.

Empirical studies show that false information spreads more quickly in its initial stages compared to true information. Using a large-scale dataset of Twitter cascades from 2006 to 2017, Vosoughi et al. (2018a) reconstruct the temporal progression of diffusion and show that false information reaches large audiences significantly faster in the early stages of propagation. This early acceleration plays a critical role in shaping overall diffusion outcomes, as misleading content can gain substantial visibility before corrective information emerges.

Such early-stage dynamics are closely linked to competition for user attention. Analysing tweet streams and user exposure patterns on Twitter, Weng et al. (2012) model how multiple pieces of information compete within users' limited attention

spans, showing that information must capture attention within a short time window to spread successfully. This highlights that early diffusion is not only faster, but also more decisive.

Beyond initial spread, misinformation diffusion often follows identifiable lifecycle patterns. Studies on rumor propagation on social media platforms, such as [Zhao et al. \(2020\)](#), analyse temporal activity sequences of rumor cascades and identify distinct phases of emergence, peak activity, and decline. Similarly, [Lerman & Ghosh \(2010\)](#), using data from Digg and Twitter, show that news diffusion typically exhibits rapid early growth followed by a sharp drop in user engagement.

Information diffusion often occurs unevenly, with sudden bursts of attention and interaction followed by quieter periods of activity. By analysing time-series data of online content popularity, including YouTube viewing activity, [Crane & Sornette \(2008\)](#) demonstrate that online attention frequently concentrates in short periods of intense activity followed by relative inactivity, rather than evolving as a smooth or continuous process. Moreover, these diffusion patterns are not driven solely by internal network processes. [Myers & Leskovec \(2014\)](#) model the interaction between internal network spread and external influences, showing that exogenous events and time-varying factors significantly shape diffusion trajectories.

The timing and evolution of information diffusion also strongly influence how effective detection and intervention strategies can be. Based on the FakeNewsNet dataset, which aggregates data from platforms such as Twitter and fact-checking sources including PolitiFact and GossipCop, [Shu et al. \(2020\)](#) show that early-stage signals in diffusion are critical for detection, as interventions become significantly less effective once misinformation has widely propagated.

While these findings highlight multiple temporal mechanisms, they can be more clearly understood when organised into distinct stages of diffusion. To provide a structured overview, Table 3.3 summarises the key characteristics of each stage, along with the role of external influences and the implications for detection and intervention.

Taken together, this stage-based perspective highlights that misinformation diffusion is not a uniform temporal process, but one that evolves through distinct phases with different dynamics, influences, and intervention opportunities. Understanding when information spreads is therefore as important as understanding how it spreads, providing a critical complement to structural analyses and a foundation for examining behavioural and platform-driven amplification in subsequent chapters.

3.4.4 Contextual Amplification in Crisis Environments

In contrast to the structural and temporal dynamics discussed in previous sections, contextual factors shape not how information diffuses, but when and to what extent it becomes amplified. In particular, crisis environments such as pandemics, public health emergencies, and rapidly evolving events create conditions under which misinformation can spread more easily and exert stronger social effects.

One defining feature of crisis contexts is the coexistence of urgency, uncertainty, and heightened demand for information. This phenomenon was widely described during the COVID-19 pandemic as an “infodemic.” Using large-scale cross-platform datasets from Twitter, Facebook, Reddit, Instagram, and YouTube, [Cinelli et al. \(2020\)](#) analyse user interactions and information flows across different online environments. Through comparative network and engagement analysis, they show that information ecosystems during crises become highly fragmented and saturated,

Table 3.3: Temporal dynamics of misinformation diffusion across different stages

Stage	Key Characteristics	External Influence	Implications for Detection and Intervention
Early Stage	Rapid growth and high diffusion speed; strong competition for user attention; initial visibility determines cascade potential	Highly sensitive to external triggers such as breaking events and algorithmic amplification	Critical stage for early detection ; intervention is most effective before large-scale spread
Peak Stage	Maximum user engagement and visibility; widespread propagation across the network; high activity levels	Amplified by trending mechanisms and collective attention dynamics	Detection becomes more difficult; misinformation already widely disseminated
Decay Stage	Declining engagement and reduced diffusion activity; attention shifts to new content	Limited external influence; driven by loss of novelty and user fatigue	Intervention has limited impact; focus shifts to mitigation and correction
Bursty Dynamics (Cross-stage)	Irregular spikes of activity interrupting smooth diffusion; non-linear temporal patterns	Often triggered by external events or re-sharing waves	Detection must account for sudden reactivation of diffusion

with users exposed to large volumes of heterogeneous and often conflicting content. Such information overload makes it increasingly difficult for individuals to distinguish reliable information from misleading claims, thereby increasing the visibility and circulation of misinformation. This broader pattern is further supported by systematic reviews of COVID-19 misinformation research, which highlight the scale and complexity of the information environment during crisis periods (Kisa & Kisa 2024).

A second contextual factor is the absence, delay, or instability of authoritative information. This dynamic is empirically demonstrated by Islam et al. (2021), who compile a large global dataset of COVID-19-related rumours from fact-checking organisations, news reports, and social media platforms. Through systematic content analysis, they classify misinformation according to its sources, themes, and potential impacts, and examine how such content circulates under different informational conditions. Their findings show that misinformation frequently emerges and spreads in contexts where authoritative knowledge is incomplete or evolving, often providing simplified or emotionally resonant explanations that fill interpretive gaps. This pattern is particularly visible in vaccine-related rumours and conspiracy theories, where uncertainty and distrust create favourable conditions for the persistence and amplification of misleading claims.

Crisis contexts are also significant because misinformation in these settings can have consequences beyond information quality alone. Drawing on a broad range of empirical studies on COVID-19, Caceres et al. (2022) analyse how misinformation affects public perception and behavioural responses to health measures. Their findings indicate that exposure to misleading information can influence risk perception, reduce compliance with public health interventions, and contribute to phenomena such as vaccine hesitancy. This suggests that misinformation diffusion in crisis contexts is not merely a communicative issue, but one with direct behavioural and societal implications.

From a broader perspective, these dynamics indicate that misinformation in crisis environments should be understood as a socio-technical problem. Analyses of information ecosystems during crises emphasise that platform dynamics and user behaviour interact in shaping diffusion outcomes. In particular, engagement-driven ranking and recommendation systems can prioritise content that attracts attention, thereby increasing the visibility of emotionally salient or controversial information during high-attention periods. At the same time, users engage under conditions of uncertainty, emotional stress, and incomplete knowledge (Denniss & Lindberg 2025). As a result, misinformation becomes not only more visible, but also more influential.

Taken together, these studies suggest that crisis environments amplify misinformation diffusion through three interacting mechanisms: (1) heightened information uncertainty, (2) gaps or instability in authoritative knowledge, and (3) increased emotional and cognitive pressure on users. These factors jointly increase the visibility, uptake, and impact of misleading information. However, amplification is not universal. Under conditions where authoritative information is timely, trusted, and widely disseminated, or where platform interventions effectively reduce exposure to low-credibility content, the spread of misinformation may be attenuated rather than amplified. In this sense, contextual amplification interacts with the structural pathways of diffusion and the temporal dynamics of attention, jointly shaping when, where, and how misinformation spreads. Understanding this contextual dimension is therefore essential for explaining why misinformation spreads unevenly across situ-

ations and why detection and intervention strategies must be adapted to the specific environments in which diffusion occurs.

3.5 What Diffusion Reveals About Misinformation?

Overall, the theoretical models and empirical findings discussed throughout this chapter suggest that misinformation and other forms of information both diffuse through communication systems. However, empirical studies consistently indicate that misinformation often exhibits distinctive diffusion characteristics when compared with other forms of information. Rather than spreading uniformly, misinformation frequently displays different patterns of propagation, influence, and amplification across communication environments.

These characteristics emerge across multiple dimensions of diffusion. As summarised in Table 3.4, differences can be observed in overall propagation outcomes, cascade structures, temporal behaviour, and contextual amplification. Together, these dimensions reveal recurring patterns through which misinformation becomes distinguishable within communication systems.

Table 3.4: Diffusion-based characteristics through which misinformation becomes observable

Aspect	What It Describes	Key Findings	Implications
Large-scale Patterns	Overall diffusion outcomes across platforms	Misinformation spreads faster, wider, and deeper, though large cascades are relatively rare	Diffusion is uneven and influenced by content, users, and platforms
Structural Characteristics	Network structure of information cascades	Diffusion forms deep, branching, and clustered cascades shaped by topology and community structure	Structure influences reach but does not fully explain diffusion variability
Temporal Dynamics	How diffusion unfolds over time	Rapid early growth, attention competition, lifecycle patterns, and bursty activity	Timing strongly shapes diffusion outcomes and intervention opportunities
Contextual Amplification	Conditions under which diffusion intensifies	Crisis environments amplify misinformation through uncertainty, authority gaps, and emotional pressure	Diffusion depends heavily on contextual and environmental conditions

From this perspective, diffusion provides more than an explanation of how information spreads. The diffusion characteristics summarised above suggest that misinformation may become visible through the patterns it produces within communication systems. Diffusion therefore offers an important perspective for understanding misinformation and provides a useful observational entry point for examining misinformation within contemporary communication environments.

Chapter 4

Understanding Misinformation Through Human Behaviour

Chapter 3 showed that misinformation often exhibits distinctive diffusion characteristics within communication systems. However, identifying these diffusion patterns does not fully explain why they emerge. If misinformation diffuses differently from other forms of information, an important question follows: what processes contribute to these differences?

To address this question, one useful approach is to examine the behaviour of actors within communication systems. This chapter examines how human behaviour has been understood within existing literature. By exploring psychological, cognitive, and social mechanisms, the chapter investigates how individuals engage with information and how these interactions contribute to misinformation diffusion. In doing so, it provides a behavioural perspective for understanding misinformation and helps explain the diffusion patterns identified in the previous chapter.

4.1 Psychological Drivers of Misinformation Sharing

Understanding why individuals share misinformation is essential for explaining its widespread diffusion in digital environments. While structural and algorithmic factors influence information visibility, a growing body of research highlights the importance of psychological drivers that shape individual sharing behaviour. Rather than acting as purely rational evaluators of information accuracy, individuals often share content based on emotional, cognitive, and social motivations. These factors shape how information is processed, evaluated, and transmitted across social networks. In particular, these psychological drivers help explain the micro-level processes that were identified as missing from system-level diffusion models in the previous chapter. These drivers can be broadly grouped into four categories: emotional arousal, novelty-driven attention, cognitive processing limitations, and social identity motivations.

Emotional arousal. One important driver of misinformation sharing is emotional arousal. Research in psychology and marketing has shown that content that evokes high-arousal emotions is more likely to be shared across social networks. In particular, emotions such as anger, anxiety, and excitement can significantly increase individuals' willingness to transmit information ([Berger & Milkman 2012](#)). Similarly, studies of online political communication demonstrate that emotionally charged and

moralised language enhances the diffusion of content within social networks (Brady et al. 2017). In the context of misinformation, such emotional dynamics can contribute to the rapid spread of misleading narratives, as sensational or provocative content is more likely to attract attention and engagement. These findings suggest that emotional arousal functions as an attention-amplifying mechanism that increases both the likelihood of engagement and the probability of onward sharing.

Novelty-driven attention. In addition to emotional factors, the novelty of information plays a crucial role in shaping sharing behaviour. Psychological theories of curiosity suggest that individuals are motivated to seek and share information that is surprising or unexpected (Loewenstein 1994). Empirical research on misinformation diffusion further supports this perspective, showing that false news tends to be more novel than accurate information and is therefore more likely to be shared (Vosoughi et al. 2018a). This tendency indicates that novelty-driven curiosity can act as a key driver in the spread of misinformation, as individuals are drawn to information that appears new or distinctive, regardless of its accuracy. In this sense, novelty operates as a selection mechanism that increases the likelihood that certain types of information are noticed and propagated within crowded information environments.

Cognitive processing limitations. Cognitive limitations also contribute significantly to the spread of misinformation. A growing body of research suggests that individuals often fail to engage in analytical reasoning when evaluating information online. Rather than carefully assessing accuracy, people frequently rely on intuitive judgments or heuristic processing. Experimental studies demonstrate that susceptibility to misinformation is better explained by a lack of cognitive reflection than by strong ideological bias (Pennycook & Rand 2019). Similarly, broader reviews of misinformation research emphasise that individuals may share misleading content not because they fully believe it, but because they do not actively consider its accuracy at the moment of sharing (Pennycook & Rand 2021). This indicates that misinformation often spreads not because of strong belief, but because individuals fail to engage in sufficient cognitive evaluation during information processing.

Social identity motivations. Finally, social and identity-related motivations further shape misinformation sharing behaviour. Individuals often use information sharing as a means of expressing identity, signalling group membership, or reinforcing social relationships. In online environments, sharing content that aligns with group norms or ideological positions may be prioritised over concerns about factual accuracy. Research on misinformation in the “post-truth” era highlights that information is frequently evaluated in relation to identity and belief systems rather than objective truth (Lewandowsky et al. 2017). As a result, misinformation may spread not only because it is persuasive or engaging, but also because it serves important social and communicative functions within groups. Here, sharing behaviour is shaped less by accuracy considerations and more by the social functions that information performs within communities.

To summarise these psychological drivers, Table 4.1 outlines the key mechanisms through which individual-level processes contribute to misinformation sharing.

Taken together, these psychological mechanisms demonstrate that misinformation sharing is not simply the result of informational deficits or lack of access to accurate data. Instead, it reflects a complex interplay of emotional responses, cognitive processes, and social motivations. Understanding these human factors is therefore essential for explaining not only why misinformation is shared at the individual level, but also how these behaviours collectively give rise to the large-scale diffusion patterns observed in digital communication systems.

Table 4.1: Psychological mechanisms underlying misinformation sharing

Mechanism	Driver	How It Works	Effect on Diffusion
Attention amplification	Emotional Arousal	High-arousal emotions increase engagement and sharing likelihood	Accelerates spread and increases virality
Information selection	Novelty-driven attention	Novel or surprising content attracts attention and is prioritised for sharing	Increases probability of diffusion
Reduced analytical processing	Cognitive processing limitations	Individuals rely on heuristics and fail to evaluate accuracy	Enables uncritical sharing of misinformation
Identity signalling	Social identity motivations	Information is shared to reinforce group belonging and social ties	Sustains diffusion within communities

4.2 Human Judgment and Truth Discernment

While the previous section examined the psychological drivers that motivate individuals to share misinformation, it is equally important to consider how individuals evaluate the accuracy of information. A common assumption is that misinformation spreads primarily because individuals are unable to distinguish between true and false information. However, research in cognitive psychology suggests that the issue is more complex. Rather than consistently engaging in deliberate and rational evaluation, individuals often rely on heuristic processing¹, prior beliefs, and contextual cues when forming judgments about information. This tendency can be understood through dual-process theories of cognition, which distinguish between fast, intuitive processing (often referred to as System 1) and slower, analytical reasoning (System 2) (Evans 2008). In particular, these mechanisms shape not only whether individuals identify misinformation, but whether they evaluate accuracy at all.

One important finding is that individuals possess a limited ability to accurately distinguish between true and false information, particularly in fast-paced digital environments. Experimental studies show that when explicitly prompted to assess accuracy, individuals can often identify false information. However, in naturalistic online settings, where information is consumed rapidly and attention is limited, individuals frequently fail to consider accuracy at all (Pennycook & Rand 2021). This indicates that misinformation often spreads not because individuals are incapable of distinguishing truth from falsehood, but because they do not engage in sufficient cognitive evaluation during information processing.

Beyond these cognitive limitations, individuals are also inclined to accept information that aligns with their prior beliefs and attitudes. Research on motivated reasoning suggests that individuals do not process information in a neutral manner, but instead selectively interpret evidence in ways that reinforce existing views (Kunda 1990). This tendency is closely related to confirmation bias, a well-documented psychological phenomenon in which individuals favour belief-consistent information while discounting contradictory evidence (Nickerson 1998). As a result, misinformation that resonates with existing beliefs is more likely to be accepted and less likely to be critically evaluated. This reflects a belief-consistent filtering process

¹Heuristic processing refers to relying on intuitive mental shortcuts rather than careful analytical reasoning when evaluating information.

that systematically biases truth judgments.

In addition, patterns of information exposure further shape truth judgments. Psychological research has demonstrated that repeated exposure to a statement increases its perceived truthfulness, a phenomenon known as the illusory truth effect (Hasher et al. 1977). In digital environments, where individuals encounter similar narratives across multiple sources and over time, repeated exposure can reinforce perceived credibility. Under such conditions, individuals may come to accept misinformation not because it has been verified, but because it feels familiar. This suggests that familiarity operates as a cognitive shortcut that substitutes for evidence in truth evaluation.

To summarise these cognitive processes, Table 4.2 outlines the key mechanisms through which human judgment is systematically biased in digital information environments.

Table 4.2: Cognitive mechanisms shaping truth judgment

Mechanism		Cognitive Process	How It Biases Judgment	Implication for Misinformation
Heuristic processing	Pro-	Fast, intuitive (System 1) thinking	Individuals rely on shortcuts rather than analytical reasoning	Reduces accuracy of truth evaluation in fast-paced environments
Inattention to Accuracy	to	Limited attention and cognitive effort	Individuals fail to consider accuracy during information consumption	Leads to passive acceptance and sharing of misinformation
Motivated Reasoning	Rea-	Belief-consistent processing	Individuals interpret information in ways that reinforce prior beliefs	Increases acceptance of ideologically aligned misinformation
Illusory Truth Effect	Truth	Familiarity-based judgment	Repeated exposure increases perceived truthfulness	Reinforces credibility of misinformation over time

Taken together, these findings indicate that misinformation persists not only because individuals lack the ability to distinguish truth from falsehood, but because human judgment is systematically shaped by cognitive mechanisms that bias information evaluation. These mechanisms do not operate in isolation; rather, they interact with broader social and group dynamics that influence how information is interpreted and shared. The next section therefore examines how social identity and political polarization further structure these processes at the collective level.

4.3 Social Structures and Political Polarization

While the previous section examined how individual cognitive processes shape the evaluation of information, these processes do not occur in isolation. In digital communication environments, individuals are embedded within social groups and networks that influence how information is interpreted and shared. As a result, misinformation should not only be understood as a product of individual cognitive limitations, but also as a socially structured phenomenon shaped by identity and group dynamics.

A key factor in this process is social identity. According to social identity theory, as developed by Tajfel et al. (2001), individuals derive part of their self-concept from

their membership in social groups and are motivated to maintain a positive image of those groups. This motivation often leads individuals to favour information that aligns with group norms and beliefs, while rejecting information that challenges them. In the context of misinformation, individuals may therefore be more likely to accept information that reinforces their group identity, and in some cases share it, regardless of its factual accuracy (Lewandowsky et al. 2017). This suggests that identity functions as a filtering mechanism, shaping which information is accepted and prioritised within group contexts.

Building on identity-driven preferences, political polarization further structures how these biases manifest at the network level. In highly polarized environments, individuals are increasingly divided along ideological lines, and information is often interpreted through the lens of group affiliation. Barberá et al. (2015) show that users tend to cluster into ideologically aligned communities and selectively engage with like-minded content. This selective engagement reinforces prior attitudes and contributes to the formation of increasingly polarized information environments. In this way, polarization operates as a network-level sorting mechanism that concentrates like-minded users and amplifies belief-consistent exposure.

These polarized interaction patterns shape not only attitudes, but also the structure of information exposure and circulation. Research has shown that individuals interacting within like-minded communities are more likely to encounter and circulate homogeneous information, leading to the formation of echo chambers (Cinelli et al. 2021). Within such environments, misinformation can persist and circulate more easily, as repeated exposure and social validation reinforce existing beliefs. Studies of misinformation diffusion further demonstrate that misinformation often spreads within tightly connected and ideologically aligned communities, where it is less likely to be challenged by opposing views (Del Vicario et al. 2016). Here, echo chambers act as a reinforcement mechanism, increasing repeated exposure to similar content while reducing opportunities for corrective information.

To summarise these collective processes, Table 4.3 outlines the key social mechanisms through which group dynamics shape misinformation diffusion.

Table 4.3: Social mechanisms shaping misinformation diffusion

Mechanism	Social Structure	How It Operates	Implication for Misinformation
Identity Filtering	Group membership	Individuals favour information aligned with group identity	Increases acceptance of identity-consistent misinformation
Network Sorting	Polarized networks	Users cluster into ideologically aligned communities	Reinforces selective exposure
Reinforcement Loop	Echo chambers	Repeated exposure and social validation within homogeneous groups	Strengthens belief and persistence of misinformation
Reduced Correction	Limited cross-group interaction	Opposing information rarely enters the network	Prevents misinformation from being challenged

Taken together, these findings suggest that misinformation diffusion is not only driven by individual cognition, but also by collective processes rooted in identity and group alignment. These mechanisms help explain why the structural and temporal patterns observed in earlier chapters often emerge in clustered and polarized forms within digital communication systems.

4.4 Collective and Societal Impacts of Misinformation

Building on the psychological and social mechanisms discussed in the preceding sections, misinformation can be understood as part of a broader feedback system linking individual cognition, information diffusion, and societal outcomes. Rather than operating as a linear process, misinformation emerges through dynamic interactions between cognitive biases, information circulation, and collective responses. Figure 4.1 illustrates this process, showing how the consequences of misinformation can feed back into individual beliefs and behaviours, reinforcing the conditions for further diffusion.

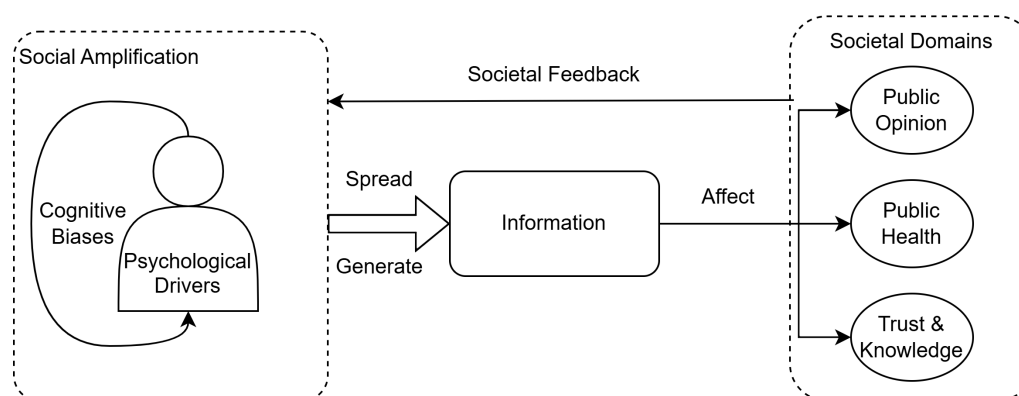


Figure 4.1: A conceptual model of misinformation as a feedback system. Psychological drivers and cognitive biases shape the generation and diffusion of information. This information produces effects across societal domains, including public discourse, public health, and trust in knowledge systems. These societal outcomes feed back into individual cognition and behaviour, reinforcing the conditions that sustain further misinformation diffusion.

One of the most widely recognised impacts of misinformation concerns democratic processes. As indicated in Figure 4.1, information shaped by identity-driven filtering and cognitive biases diffuses through socially reinforced pathways, influencing how individuals perceive political reality. This process can shape public discourse, distort political communication, and affect electoral behaviour. Research suggests that such dynamics undermine informed decision-making and weaken democratic institutions (Lazer et al. 2018). Importantly, these outcomes arise not directly from misinformation itself, but from the interaction between individual cognition and the structured diffusion of information within social systems.

Misinformation also has significant implications for public health and risk communication. As shown in Figure 4.1, the interaction between cognitive limitations—such as inattention and heuristic processing²—and large-scale information diffusion influences how individuals interpret risk-related information. During crises such as the COVID-19 pandemic, misleading information spreads rapidly through networks, shaping perceptions of risk and medical advice. Consequently, individuals may make decisions based on incomplete or inaccurate information, affecting behaviours such as vaccine uptake and adherence to public health guidelines (Caceres et al. 2022). These behavioural outcomes illustrate how individual cognition and

²Heuristic processing refers to a mode of information evaluation in which individuals rely on simple mental shortcuts or intuitive judgments rather than careful analytical reasoning.

information dynamics jointly shape real-world consequences.

Beyond specific domains such as politics and health, misinformation contributes to broader challenges related to trust and knowledge in society. Within echo chambers and homophilous networks³, repeated exposure to similar or conflicting information reinforces selective perception and limits exposure to corrective content. As represented in Figure 4.1, these structural conditions of information diffusion contribute to the erosion of confidence in authoritative sources. Scholars argue that such dynamics foster a “post-truth” environment in which individuals become increasingly uncertain about what information can be trusted (Lewandowsky et al. 2017).

In this sense, these findings suggest that misinformation should be understood as a dynamic and self-reinforcing process rather than a collection of isolated psychological, social, or societal effects. Individual cognition, information diffusion, and societal outcomes continuously influence one another, creating feedback processes that can sustain and amplify misinformation over time. Understanding these interactions is therefore important not only for explaining why misinformation persists, but also for informing efforts to monitor, mitigate, and respond to misinformation within contemporary communication environments.

4.5 What Human Behaviour Reveals About Misinformation?

Overall, the literature reviewed in this chapter highlights the importance of human behaviour in shaping how information is attended to, evaluated, shared, and reinforced within communication systems. The recurring patterns through which individuals and groups interact with information—including emotional engagement, selective interpretation, identity reinforcement, and repeated exposure—provide important insights into how misinformation operates within communication systems.

From this perspective, human behaviour offers not only an explanation for misinformation diffusion, but also an additional perspective through which misinformation can be observed and understood. Rather than becoming visible solely through patterns of diffusion, misinformation may also become observable through the ways individuals and groups interact with information within socio-technical communication environments.

³Homophilous networks refer to social networks in which individuals with similar beliefs, identities, or characteristics are more likely to connect and interact with one another.

Chapter 5

Understanding Misinformation Through Platform Dynamics

The previous chapter showed that misinformation can be understood through the ways individuals and groups interact with information. However, these interactions increasingly occur within platform environments that shape what information becomes visible and how it circulates online. This raises a further question: how do platforms influence the spread and persistence of misinformation?

To address this question, this chapter examines how platform dynamics have been understood within existing literature. It explores how visibility systems, recommendation mechanisms, and governance interventions shape information exposure, amplification, and regulation within digital environments. In doing so, it examines how platform environments shape human–information interactions and influence the diffusion of misinformation within communication systems.

5.1 Visibility Dynamics in Platform Environments

One of the most significant ways digital platforms shape online communication is through the management of informational visibility and exposure conditions. A growing body of research argues that platforms function as active visibility structures rather than neutral communication channels. [Gillespie \(2014\)](#) argues that algorithms play a central role in determining what information is considered relevant and visible within digital environments. From this perspective, platforms influence public knowledge not simply by hosting information, but by structuring exposure conditions through ranking, recommendation, and curation systems. Similarly, [Bucher \(2012\)](#), analysing Facebook’s EdgeRank system, demonstrates that algorithmic systems create conditions of uneven visibility in which users adapt their behaviour in response to perceived risks of becoming “invisible” within platform environments. Together, these studies suggest that visibility within digital communication systems is not distributed equally, but actively managed through platform infrastructures.

Empirical research further demonstrates how algorithmic curation reshapes patterns of information exposure. Using Twitter timeline data, [Bandy & Diakopoulos \(2021\)](#) compare algorithmically curated timelines with chronological timelines and show that algorithmic curation concentrates user attention on a narrower set of sources while reducing the diversity of encountered information. Their findings suggest that platform curation systems actively shape informational exposure conditions rather than simply reflecting user preferences. This has important implications

for misinformation diffusion, as visibility becomes increasingly tied to behavioural engagement signals rather than informational accuracy.

Figure 5.1 illustrates this shift from chronological exposure to algorithmically curated visibility. In a chronological model, exposure is primarily determined by temporal order, meaning that content visibility is assumed to be relatively neutral and sequential. By contrast, in algorithmically curated environments, content first passes through ranking and filtering systems that prioritise some information over others according to platform objectives and behavioural signals. This distinction is important because it shows that contemporary diffusion does not begin only when users decide to share content; it begins earlier, when platforms determine which information becomes visible in the first place.

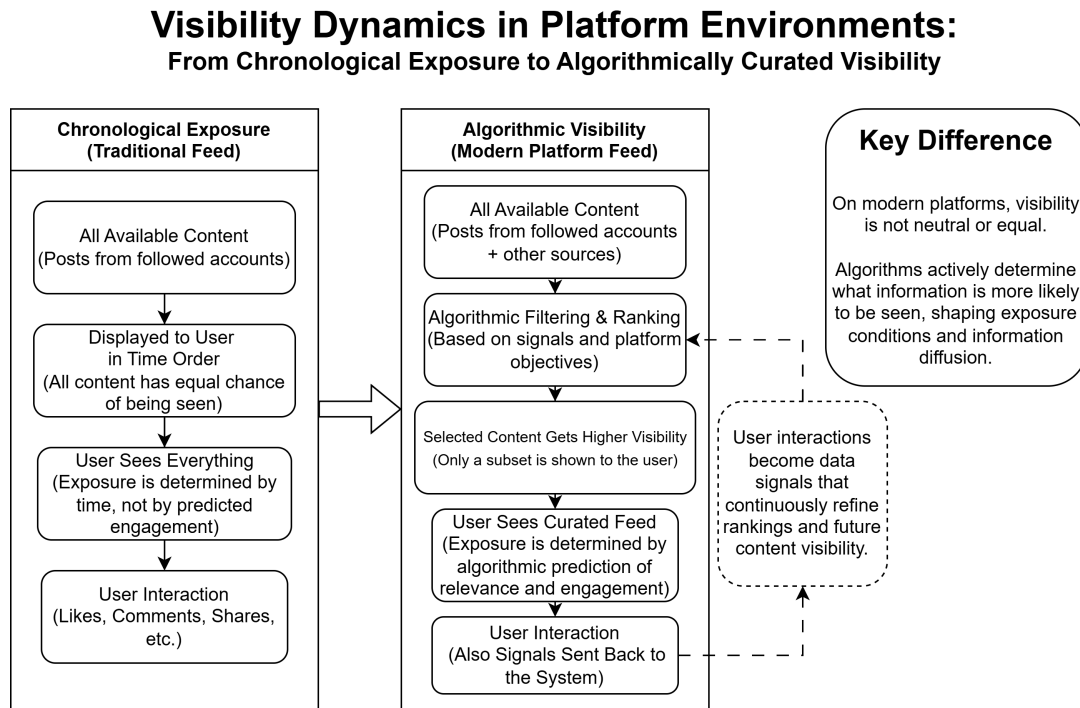


Figure 5.1: Comparison between chronological exposure and algorithmically curated visibility structures in digital platform environments.

The increasing dependence of visibility on engagement metrics further intensifies these dynamics. Engagement-based ranking systems frequently prioritise clicks, shares, watch time, comments, and interaction frequency in order to maximise user activity and retention. As a result, content that generates strong emotional or behavioural reactions may achieve disproportionate visibility regardless of factual reliability (Berger & Milkman 2012). Research on engagement-driven ranking systems demonstrates that optimization for engagement can contribute to the amplification of emotionally engaging misinformation and increased polarization within online environments (Germano et al. 2026).

Importantly, these amplification processes emerge through interactions between user behaviour and algorithmic systems. In a systematic review of digital media systems, Metzler et al. (2023) argue that misinformation diffusion is shaped by feedback loops linking human behavioural tendencies and algorithmic amplification mechanisms. From this perspective, algorithms do not independently create misinformation diffusion, but amplify behavioural signals generated through user interactions.

These dynamics challenge several assumptions underlying classical diffusion models discussed in earlier chapters. Rather than occurring under relatively stable exposure conditions, information diffusion within contemporary platform environments is increasingly shaped by algorithmic curation, engagement optimisation, and platform architectures that structure visibility and attention. As a result, platforms influence not only what information spreads, but also how users encounter and engage with information online.

5.2 Recommendation Systems and Reinforcement Dynamics

While the previous section examined how platforms structure information visibility, visibility alone does not explain why some forms of content become continuously amplified across digital environments. Contemporary social media systems increasingly rely on recommendation and engagement optimization mechanisms that reinforce behavioural signals through recursive feedback processes. As a result, misinformation diffusion is shaped not only by exposure conditions, but also by platform systems that continuously amplify content generating strong engagement.

5.2.1 Engagement as an Optimization Objective

A defining characteristic of contemporary digital platforms is the prioritisation of engagement optimization. Recommendation and ranking systems frequently rely on behavioural metrics such as clicks, watch time, likes, shares, comments, and interaction frequency in order to maximise user attention and platform activity. Within such systems, engagement effectively functions as a proxy for relevance.

This optimization logic is particularly visible in large-scale recommendation systems. Describing the architecture of YouTube's recommendation system, [Covington et al. \(2016\)](#) demonstrate how recommendation models continuously optimise user watch time and retention through personalised ranking systems trained on large-scale behavioural interaction data. Their work highlights how recommendation systems dynamically adapt visibility according to behavioural feedback signals, positioning user engagement as a central component of platform optimization processes.

The prioritisation of engagement over informational accuracy has important implications for misinformation diffusion. Research on engagement-driven ranking systems suggests that optimization for behavioural interaction can amplify emotionally engaging and polarising content regardless of factual reliability ([Germano et al. 2026](#)). As platforms increasingly reward interaction intensity, emotionally provocative or attention-grabbing misinformation may achieve disproportionate visibility within digital environments. Importantly, this amplification does not necessarily result from deliberate ideological objectives, but can emerge as an unintended consequence of optimization systems designed to maximise user activity, retention, and behavioural engagement.

5.2.2 Emotional Engagement and Algorithmic Amplification

The relationship between engagement optimization and misinformation diffusion is closely connected to the psychological mechanisms discussed in Chapter 5. Research consistently demonstrates that emotionally arousing and novel content generates stronger behavioural interaction and higher diffusion potential online.

As discussed previously, [Berger & Milkman \(2012\)](#) show that high-arousal emotional content is significantly more likely to be shared online, while [Brady et al. \(2017\)](#) further demonstrate that moral-emotional language substantially increases online diffusion within social networks. Similarly, [Vosoughi et al. \(2018a\)](#) find that false news spreads more rapidly and broadly than truthful information, partly because it generates stronger novelty and engagement signals.

Within contemporary platform systems, these behavioural tendencies become increasingly significant because algorithms interpret user interaction as indicators of relevance and popularity. Emotional engagement therefore functions not only as a psychological driver of sharing behaviour, but also as an algorithmically amplified behavioural signal. As a result, misinformation that evokes strong emotional reactions may receive increased visibility through recommendation and ranking systems, creating conditions for large-scale amplification. These dynamics illustrate that platform amplification is closely linked to human behaviour: recommendation systems do not independently generate misinformation diffusion, but intensify existing engagement patterns by rewarding the forms of content most likely to sustain user attention and interaction.

5.2.3 Recursive Feedback Loops

The interaction between behavioural engagement and recommendation systems produces recursive feedback loops that play a central role in contemporary misinformation diffusion. These feedback dynamics emerge when user engagement influences algorithmic visibility, which subsequently reshapes future exposure conditions and behavioural responses.

In a systematic review of research on digital media systems, [Metzler et al. \(2023\)](#) examine how platform algorithms, behavioural interaction, and social dynamics jointly shape online information diffusion. Their analysis highlights how engagement-based algorithms continuously reinforce behavioural signals generated through user interaction, producing recursive cycles of visibility and engagement.

Research on online communities further demonstrates how such reinforcement dynamics operate within social networks. As discussed in Chapter 5, [Cinelli et al. \(2021\)](#) show that users increasingly cluster into ideologically and behaviourally homogeneous communities, reinforcing repeated exposure to similar forms of information. Likewise, [Del Vicario et al. \(2016\)](#) demonstrate that misinformation tends to diffuse within highly aligned communities where repeated interaction and social reinforcement strengthen informational persistence.

Together, these studies suggest that misinformation amplification is not simply a linear process of transmission, but a recursive system in which engagement generates visibility, visibility produces further interaction, and interaction subsequently reinforces future exposure conditions. Amplification therefore emerges through continuous feedback between users, algorithms, and social network structures.

Figure 5.2 illustrates these recursive amplification dynamics within digital plat-

form environments. Rather than operating as a simple linear recommendation process, misinformation amplification emerges through continuous feedback between behavioural engagement, algorithmic visibility systems, and exposure conditions. User interactions generate behavioural signals that influence future visibility allocation, while repeated exposure further reinforces subsequent engagement and interaction. Over time, such recursive dynamics may contribute to broader systemic consequences including the rapid amplification of misinformation, increasingly homogeneous and polarized exposure environments, and more fragmented and emotionally intensified forms of public discourse within digital communication systems.

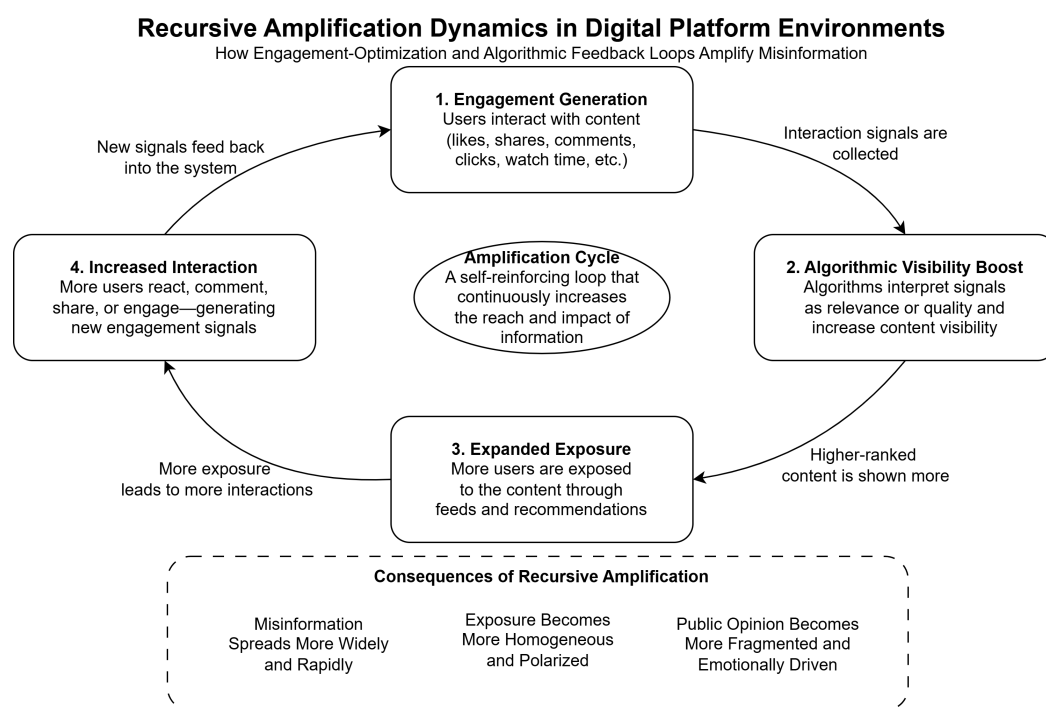


Figure 5.2: Recursive amplification dynamics in digital platform environments.

From a system perspective, these recursive amplification processes can be understood as feedback systems in which behavioural outputs continuously become informational inputs for future algorithmic adaptation. Platforms monitor engagement, adjust visibility conditions accordingly, and thereby reproduce the behavioural patterns most likely to sustain interaction.

5.2.4 Adaptive Amplification in Platform Environments

Beyond recursive reinforcement alone, recommendation systems also continuously adapt exposure conditions according to behavioural feedback. As platforms collect increasingly large volumes of interaction data, recommendation systems dynamically modify visibility pathways in response to user behaviour, attention patterns, and engagement histories.

Research on YouTube recommendation systems illustrates this adaptive process particularly clearly. Using large-scale YouTube recommendation graph data, [Ribeiro et al. \(2020\)](#) analyse recommendation trajectories in order to examine how users may be guided toward increasingly homogeneous or extreme content environments. Through recommendation pathway analysis and graph modelling, the authors demonstrate that recommendation systems can progressively direct users

toward highly engaging and ideologically aligned content through sequential recommendation chains.

Experimental research on social media feed algorithms further demonstrates how algorithmic systems dynamically restructure exposure conditions. Using large-scale field experiments conducted on Facebook and Instagram during the 2020 United States presidential election, [Guess et al. \(2023\)](#) compare algorithmically curated feeds with chronological feeds in order to examine how ranking systems shape user exposure and engagement. Their findings demonstrate that algorithmic feeds significantly alter informational exposure and behavioural interaction patterns relative to chronological systems, providing empirical evidence that platform algorithms actively structure exposure environments.

These findings highlight that misinformation amplification within digital platforms is not static, but adaptive and continuously evolving. Recommendation systems dynamically reshape informational environments in response to behavioural feedback signals, creating changing patterns of visibility and interaction. As a result, recommendation systems do not merely distribute information across networks, but actively reinforce and amplify behavioural signals through recursive optimisation processes, contributing to the large-scale diffusion and persistence of misinformation within contemporary digital environments.

5.3 Platform Governance and Moderation Dynamics

An important question surrounding platform governance is whether moderation systems are actually effective in reducing misinformation diffusion. This question has become increasingly significant as digital platforms expand their use of content removal, algorithmic downranking¹, account restrictions, fact-check labels, and automated moderation systems to manage misleading information online.

However, evaluating the effectiveness of platform governance is not straightforward. Different moderation strategies often pursue different objectives, including reducing visibility, limiting amplification, disrupting coordination, or discouraging harmful behaviour. At the same time, because digital communication environments remain adaptive and interconnected, governance interventions may also generate unintended consequences such as behavioural adaptation, migration across platforms, and changes in user interaction patterns.

For example, investigating the effects of deplatforming across social media environments, [Horta Ribeiro et al. \(2025\)](#) conduct a longitudinal quasi-experimental analysis of 165 deplatforming events targeting 101 online influencers. Combining Reddit records of deplatforming events with Google Trends and Wikipedia page-view data, the study finds that deplatforming significantly reduced overall online attention toward targeted influencers across multiple platforms over time. These findings suggest that governance interventions may successfully reduce misinformation visibility and online attention under certain conditions, although their broader systemic effects remain more complex and difficult to evaluate.

The following sections therefore examine different forms of platform governance and moderation in greater detail, including visibility regulation, automated moderation systems, and the unintended consequences and adaptive responses generated by

¹Algorithmic downranking refers to reducing the visibility or recommendation priority of content within platform ranking systems without fully removing the content itself.

governance interventions within contemporary digital communication environments.

5.3.1 Content Moderation and Visibility Control

While previous sections examined how platforms algorithmically allocate visibility in order to optimise engagement and exposure, platform governance systems increasingly attempt to regulate visibility in order to manage misinformation. Through labeling systems, content removal policies, algorithmic downranking, and account restrictions, platforms attempt to reduce user exposure to misleading content and limit its diffusion potential.

Research on platform governance increasingly emphasises that moderation systems function as mechanisms of visibility control rather than simple forms of content deletion. In *Custodians of the Internet*, Gillespie (2018) argue that digital platforms increasingly operate as “custodians” of online discourse, actively shaping what content remains visible, amplified, or suppressed within digital communication environments. Rather than functioning as neutral intermediaries, platforms continuously structure informational visibility through moderation policies and algorithmic governance systems.

Similarly, Klonick (2017) argue that major social media companies increasingly function as private governors of online speech. Through legal and institutional analysis of platform policies and moderation systems, the study demonstrates that platforms exercise substantial authority over digital communication through internally defined rules, enforcement systems, and visibility management practices.

More broadly, Gorwa (2019) conceptualise platform governance as a complex interaction between technological systems, corporate policies, state regulation, and user practices. Reviewing the broader governance literature surrounding digital platforms, the study argues that platform governance extends beyond individual moderation decisions and instead involves the management of socio-technical communication infrastructures. This perspective highlights that moderation systems operate not simply as technical filtering tools, but as governance mechanisms shaping the structure of digital public discourse.

These dynamics are particularly significant because visibility itself plays a central role in diffusion processes. As discussed in previous sections, recommendation and ranking systems determine exposure conditions for users. Consequently, moderation interventions that reduce visibility may significantly alter the trajectories through which misinformation spreads across digital environments.

At the same time, visibility regulation introduces important tensions regarding transparency, legitimacy, and platform power. Because moderation decisions influence what users encounter online, platforms increasingly occupy positions of informational governance traditionally associated with editors, broadcasters, or public institutions. Debates surrounding misinformation moderation therefore extend beyond technical concerns into broader political and societal questions regarding authority, censorship, and control over digital communication environments.

5.3.2 Algorithmic Moderation and Automated Detection

As the scale of online communication continues to expand, platforms increasingly rely on automated moderation systems to identify and manage harmful or misleading content. Machine learning models, natural language processing systems, and automated detection algorithms are now widely used to detect misinformation, hate

speech, coordinated manipulation, and other forms of problematic content at scale.

Automated moderation systems provide important advantages in terms of scalability and speed. However, research also highlights significant limitations associated with algorithmic governance. In *Behind the Screen*, Roberts (2019) examine the operational structures of commercial content moderation systems and demonstrate that moderation remains deeply dependent on contextual interpretation, human judgement, and platform-specific decision-making processes. The study highlights the limitations of fully automated moderation systems, particularly when content meaning depends heavily on context, culture, sarcasm, or rapidly evolving narratives.

Research on human-machine collaboration further reinforces these limitations. Analysing Reddit's AutoModerator system, Jhaver et al. (2019) examine how automated moderation tools interact with human moderators within large online communities. Using Reddit moderation records and governance workflow analysis, the authors demonstrate that automated moderation systems require continuous human adjustment and contextual interpretation in order to remain effective. Their findings suggest that algorithmic moderation functions less as a fully autonomous system and more as an adaptive socio-technical process involving ongoing interaction between automated tools and human governance practices.

These limitations become particularly significant in misinformation environments because misinformation itself is dynamic and adaptive. Narratives, symbols, and communication strategies frequently evolve in response to moderation systems, reducing the effectiveness of static detection approaches. As a result, moderation systems must continuously adapt to changing forms of informational manipulation and user behaviour. From a socio-technical perspective, moderation can therefore be understood as an ongoing process of system regulation rather than a fixed technical solution, in which platforms continuously monitor behavioural and informational signals, adjust governance mechanisms, and refine detection systems in response to changing diffusion dynamics.

5.3.3 Unintended Consequences and Adaptive Responses

Moderation interventions may also generate unintended consequences that reshape diffusion dynamics in unexpected ways. Because misinformation ecosystems are adaptive, interventions designed to suppress harmful content may alter user behaviour, platform migration patterns, and perceptions of institutional legitimacy.

Research on online moderation demonstrates that users frequently adapt to governance interventions rather than simply ceasing participation. Examining Reddit's 2015 ban on hate-related communities, Chandrasekharan et al. (2017) analyse behavioural changes before and after subreddit bans using large-scale Reddit activity data. Through comparative analysis of user activity patterns, the study finds that banning highly toxic communities significantly reduced hate speech participation within Reddit overall. However, the authors also observe adaptive responses, including user migration² and attempts to reconstruct banned communities elsewhere on the platform.

Research on media manipulation and disinformation further demonstrates that actors spreading misleading information frequently adapt their communication strate-

²User migration refers to the movement of users or online communities from one platform or digital environment to another, often in response to moderation interventions, governance restrictions, or changes in visibility conditions.

gies in response to platform governance systems. [Marwick & Lewis \(2017\)](#) analyse online disinformation and media manipulation practices within contemporary digital environments, demonstrating how coordinated actors strategically exploit platform architectures, algorithmic systems, and communication norms in order to maximise visibility and evade governance interventions. Such adaptive strategies may include coded language, indirect messaging, coordinated amplification, or the modification of visual and textual formats to avoid automated detection systems.

Moderation pressures may also contribute to migration toward alternative or less regulated platforms. Investigating the movement of controversial online communities toward Telegram and alternative social media environments, [Rogers \(2020\)](#) use cross-platform tracing and migration analysis to examine how users respond to deplatforming³ interventions. Their findings suggest that deplatforming does not necessarily eliminate problematic communities, but may instead redistribute them across different digital ecosystems with weaker moderation structures.

Table 5.1 summarises common adaptive responses to platform governance interventions. The examples illustrate how moderation measures often generate behavioural and systemic adjustments within digital communication environments. Rather than functioning as static forms of external control, governance interventions become embedded within ongoing interactions among users, algorithms, platform infrastructures, and evolving communication practices.

Table 5.1: Governance interventions and adaptive responses in digital platform environments

Governance Intervention	Intended Function	Adaptive or Systemic Response
Content removal and account suspension	Reduce visibility of harmful or misleading content	User migration and reconstruction of communities on alternative platforms
Algorithmic down-ranking and visibility reduction	Limit exposure without fully removing content	Reduced transparency and difficulties in identifying moderation practices
Automated moderation systems	Enable scalable detection and filtering of problematic content	Contextual limitations, false positives, and dependence on human oversight
Fact-check labels and informational warnings	Increase awareness of misinformation and discourage sharing	Strategic adaptation, coded language, and evasion strategies
Deplatforming and community bans	Disrupt coordinated harmful activity	Redistribution of users across weaker or less regulated ecosystems

Taken together, these developments suggest that moderation systems operate within complex adaptive environments shaped by continuous interactions between users, algorithms, platforms, and informational behaviour. Platform interventions influence visibility conditions, behavioural incentives, and diffusion pathways, while users and misinformation ecosystems continuously adapt in response. As a result,

³Deplatforming refers to the removal or restriction of individuals, groups, or communities from digital platforms in order to limit their visibility and participation online.

governance interventions do more than simply suppress misinformation; they actively reshape the informational environments within which misinformation emerges, circulates, and evolves. Understanding these dynamics is therefore important not only for platform governance, but also for understanding how misinformation can be observed, detected, and analysed within contemporary digital communication systems.

5.4 What Platform Dynamics Contribute to Understanding Misinformation?

Overall, the literature reviewed in this chapter suggests that platforms do more than simply host or distribute information. Through visibility management, recommendation systems, and governance interventions, platforms actively shape the conditions under which information is encountered, amplified, and regulated within digital communication environments.

From this perspective, platform dynamics provide an important environmental context for understanding misinformation. Rather than determining misinformation directly, platform architectures shape the visibility conditions, behavioural incentives, and governance structures within which misinformation diffusion and human–information interactions occur. As a result, platforms play a central role in influencing how misinformation spreads, persists, and adapts within contemporary communication systems.

Chapter 6

Detection Approach of Misinformation

As discussed from chapter 3 to chapter 5, misinformation can be understood through its diffusion characteristics and patterns of human–information interaction, both of which are shaped by broader socio-technical environments. These perspectives reveal different ways in which misinformation becomes observable within communication systems. In existing research, computational detection has become one of the main ways of operationalising such observation, by translating observable characteristics of misinformation into measurable indicators and classification tasks.

This chapter therefore examines how misinformation has been operationalised through major detection paradigms. It discusses content-based approaches that analyse informational content, propagation-based approaches that examine diffusion dynamics, and behavioural approaches that focus on user interaction patterns. In doing so, the chapter considers both what these methods make observable and why they remain limited within increasingly adaptive digital communication environments.

6.1 Content-Based Approaches

Content-based detection approaches represent one of the earliest and most widely used methods for identifying misinformation in digital communication environments. Rather than examining how information spreads or how users interact with it, these approaches focus on the information itself. The basic assumption is that misinformation may exhibit distinctive textual, semantic, stylistic, or visual characteristics that can be identified computationally. As a result, content is transformed into machine-readable representations and analysed using machine learning and natural language processing techniques to determine whether it is likely to be misleading or false. Over time, these approaches have evolved from simple feature-based methods to more sophisticated systems incorporating contextual language models, multimodal analysis, and large-scale pretrained models.

Early research on misinformation and credibility assessment focused primarily on manually engineered linguistic and stylistic features. Using Twitter credibility analysis, [Castillo et al. \(2011\)](#) demonstrate that features related to message content, writing style, and linguistic structure can contribute to distinguishing credible from non-credible information online. Similarly, [Rubin \(2017\)](#) examine deceptive and misleading forms of online communication, highlighting how linguistic cues, rhetorical structures, and stylistic patterns may reveal characteristics associated

with misinformation, satire, or deceptive narratives. These early studies established the foundational assumption that misinformation may contain detectable textual signals that differ from more reliable forms of information.

As misinformation research expanded, content-based detection increasingly incorporated machine learning approaches designed to automate large-scale classification tasks. Reviewing the broader development of misinformation detection research, [Shu et al. \(2017b\)](#) identify content analysis as one of the central paradigms within fake news detection systems. Such approaches commonly rely on textual representations, semantic features, lexical patterns, sentiment analysis, readability measures, and syntactic characteristics in order to classify information as credible or misleading. During this stage, classical machine learning models including support vector machines, naive Bayes classifiers, and random forest models became widely used for misinformation classification tasks.

The development of benchmark datasets further accelerated the growth of automated content-based detection research. One influential example is the LIAR dataset introduced by [Wang \(2017\)](#), which provides large-scale labelled political statements for misinformation classification tasks. The increasing availability of labelled misinformation datasets enabled researchers to train and evaluate computational models across increasingly large corpora of online information. Similarly, [Shu et al. \(2020\)](#) introduce FakeNewsNet, a large-scale repository integrating news content with social context and user interaction data. Such datasets contributed significantly to the operationalisation and standardisation of misinformation detection research within computational social science and natural language processing.

As computational methods advanced, misinformation detection increasingly shifted toward deep learning and neural language models capable of capturing more complex semantic relationships within text. [Ruchansky et al. \(2017\)](#) propose hybrid deep learning models that integrate textual, temporal, and social information for misinformation detection, illustrating a broader transition toward more sophisticated machine learning architectures. More recently, transformer-based language models such as BERT ([Devlin et al. 2019](#)) further improved contextual language representation and semantic classification capabilities within content-based misinformation detection systems. Building on these developments, [Kaliyar et al. \(2021\)](#) demonstrate how transformer-based architectures can improve misinformation classification performance by capturing contextual and semantic information more effectively than earlier feature-engineered approaches.

At the same time, misinformation environments became increasingly multimodal. Rather than circulating solely through textual content, contemporary misinformation frequently incorporates images, videos, memes, synthetic audio, and cross-platform visual communication. This development expanded content-oriented detection systems beyond purely textual analysis toward multimodal detection approaches capable of integrating multiple forms of information simultaneously. Reviewing the broader development of multimodal machine learning, [Baltrušaitis et al. \(2018\)](#) demonstrate how combining multiple modalities can improve computational understanding of complex communication environments.

As a result, researchers increasingly developed multimodal fusion methods that combine textual, visual, and contextual information within unified computational architectures. These approaches often rely on cross-modal embeddings,¹ feature fusion techniques, and multimodal transformer systems capable of jointly analysing

¹Cross-modal embeddings are computational representations that align information from different modalities, such as text and images, within a shared representational space.

visual and textual information. One influential example is CLIP,² introduced by Radford et al. (2021), which learns transferable relationships between images and natural language descriptions. Such approaches illustrate how misinformation detection increasingly attempts to model interactions across multiple communication modalities rather than analysing isolated textual signals independently.

The rapid growth of synthetic media and AI-generated misinformation further intensified these developments. Contemporary misinformation increasingly includes deepfakes,³ AI-generated images, synthetic speech, and machine-generated textual misinformation capable of closely resembling authentic communication. Reviewing the broader risks associated with deepfakes, Chesney & Citron (2019) highlight their implications for democracy, privacy, and information integrity. Similarly, Tolosana et al. (2020) survey computational approaches for detecting manipulated visual media, including convolutional neural network models, facial inconsistency analysis, and frequency-domain detection methods. More recently, Zhou et al. (2023) examine how generative AI systems increasingly produce highly coherent and persuasive misinformation that complicates conventional detection approaches.

Alongside multimodal systems, misinformation detection research also increasingly incorporated transfer learning and foundation models. Rather than training models from scratch for individual detection tasks, transfer learning approaches utilise large pretrained language or multimodal models capable of generalising across domains and communication environments. Recent developments in large language models and foundation models further expanded the possibility of few-shot and zero-shot misinformation detection across rapidly evolving information environments (Bommasani et al. 2021, Brown et al. 2020).

More recently, large language models (LLMs) have further reshaped content-based misinformation detection. Earlier systems primarily focused on classifying content according to predefined linguistic or semantic features, whereas LLM-based approaches increasingly incorporate contextual reasoning, explanation generation, and claim evaluation. For example, Hu et al. (2024) explore how large language models can support fake news detection not only as classifiers but also as reasoning agents capable of evaluating informational claims. These developments reflect a broader shift from feature-based classification towards reasoning-oriented approaches to misinformation detection.

At the same time, researchers have increasingly questioned the stability of the features used by content-based detection systems. As misinformation evolves across platforms and communication environments, models trained on particular datasets may struggle to generalise beyond the conditions under which they were developed (Wu et al. 2024). This raises broader questions about what such systems actually learn to identify and how robust these signals remain within adaptive information environments.

To summarise the evolution of content-based and multimodal misinformation detection systems, Table 6.1 outlines the major stages in the development of computational detection systems, highlighting how changing misinformation environments contributed to the adoption of increasingly sophisticated models and techniques.

These developments enabled content-based and multimodal detection systems to be applied across a wide range of practical contexts, including fake news classifi-

²CLIP (Contrastive Language-Image Pretraining) is a multimodal machine learning model designed to learn relationships between textual descriptions and visual images.

³Deepfakes are synthetic media generated or manipulated using artificial intelligence techniques to produce realistic but fabricated audio, images, or video content.

Table 6.1: Evolution of content-based and multimodal misinformation detection systems

Development Stage	Detection Focus	Models and Techniques	Misinformation Environment	Key Limitations
Early linguistic analysis	Stylistic and rhetorical credibility cues	Handcrafted linguistic and stylistic features	Primarily textual misinformation	Limited semantic understanding
Classical machine learning era	Automated content classification	Support vector machines, naive Bayes, random forest models	Rapid growth of online textual misinformation	Context dependence and shallow feature representation
Benchmark dataset expansion	Large-scale supervised misinformation classification	LIAR, FakeNewsNet, labelled misinformation corpora	Increasing scale of social media communication	Dataset bias and limited generalisability
Deep learning era	Semantic representation learning	Convolutional neural networks, recurrent neural networks, hybrid neural models	More complex online narratives and large-scale digital content	Limited explainability and high data requirements
Transformer era	Contextual language modelling	BERT, FakeBERT, transformer-based language models	High-volume and rapidly evolving online discourse	Computational complexity and contextual ambiguity
Multimodal detection era	Cross-modal misinformation analysis	CLIP, multimodal transformers, feature fusion systems	Images, videos, memes, and cross-platform communication	Cross-modal ambiguity and multimodal integration challenges
Generative AI era	Detection of synthetic and AI-generated content	Deepfake detection systems, synthetic media classifiers, foundation models	AI-generated misinformation and synthetic communication environments	Rapid adaptation, realism, and evolving generative systems
LLM reasoning era	Claim verification and reasoning-based detection	GPT-based systems, LLM judges, retrieval-augmented verification, multi-agent debate frameworks	Adaptive information environments and generative AI ecosystems	Reasoning reliability, hallucinations, explainability, and adversarial manipulation

cation, fact verification, clickbait detection, deepfake identification, and automated moderation. Because these approaches can be automated and scaled efficiently, they remain central to contemporary misinformation detection systems.

However, content-based approaches also face important limitations. Misinformation is often highly context-dependent (Wardle & Derakhshan 2017), and its influence is shaped not only by informational content but also by diffusion processes, user behaviour, and platform environments. Recent studies further suggest that detection systems may struggle when misinformation is reformulated or adapted to new contexts (Wu et al. 2024, Przybyła et al. 2025). Content-based detection therefore provides an important way of operationalising misinformation, but it captures only one dimension of a broader socio-technical phenomenon.

6.2 Network and Propagation-Based Detection

While content-based approaches attempt to identify misinformation through textual and semantic characteristics, propagation-based approaches focus on how information spreads through digital communication networks. These approaches assume that misinformation may exhibit distinctive diffusion characteristics, including particular network structures, cascade patterns, and temporal dynamics. Rather than analysing informational content directly, propagation-based detection systems transform observable diffusion behaviour into computational representations that can support misinformation inference. Figure 6.1 illustrates the general operational logic of this approach.

One important line of research focuses on cascade-based detection methods. These approaches analyse the structural properties of information diffusion cascades,⁴ including cascade depth, breadth, size, and branching structure. Research on online diffusion demonstrates that different types of information may generate distinct propagation patterns across social networks. In particular, Vosoughi et al. (2018a) show that false news spreads significantly faster, deeper, and more broadly than truthful information on Twitter. Similarly, Goel et al. (2016) introduce the concept of structural virality⁵ to analyse how information propagates through networked communication systems. These studies contributed to the broader assumption that misinformation cascades may contain detectable structural signatures that can be operationalised for classification and detection purposes.

Beyond structural characteristics, researchers also began analysing temporal propagation patterns as potential detection signals. Temporal detection approaches focus on how misinformation diffusion evolves over time, including early spread velocity, burst dynamics,⁶ lifecycle progression, and fluctuations in user engagement. Reviewing misinformation propagation research, Zhao et al. (2020) highlight how rumours and misleading information often exhibit distinctive temporal diffusion patterns across social media environments. Similarly, Crane & Sornette (2008) demonstrate that online attention frequently concentrates in short bursts of activity rather than evolving smoothly over time. Using large-scale Twitter diffusion data, Myers & Leskovec (2014) further show that diffusion trajectories are shaped not

⁴A diffusion cascade refers to the process through which information spreads from one user to another across a network, forming branching propagation structures over time.

⁵Structural virality refers to the extent to which information diffusion resembles decentralised peer-to-peer sharing rather than direct broadcast from a single central source.

⁶Burst dynamics refer to sudden periods of intense information activity and rapid diffusion followed by relative inactivity.

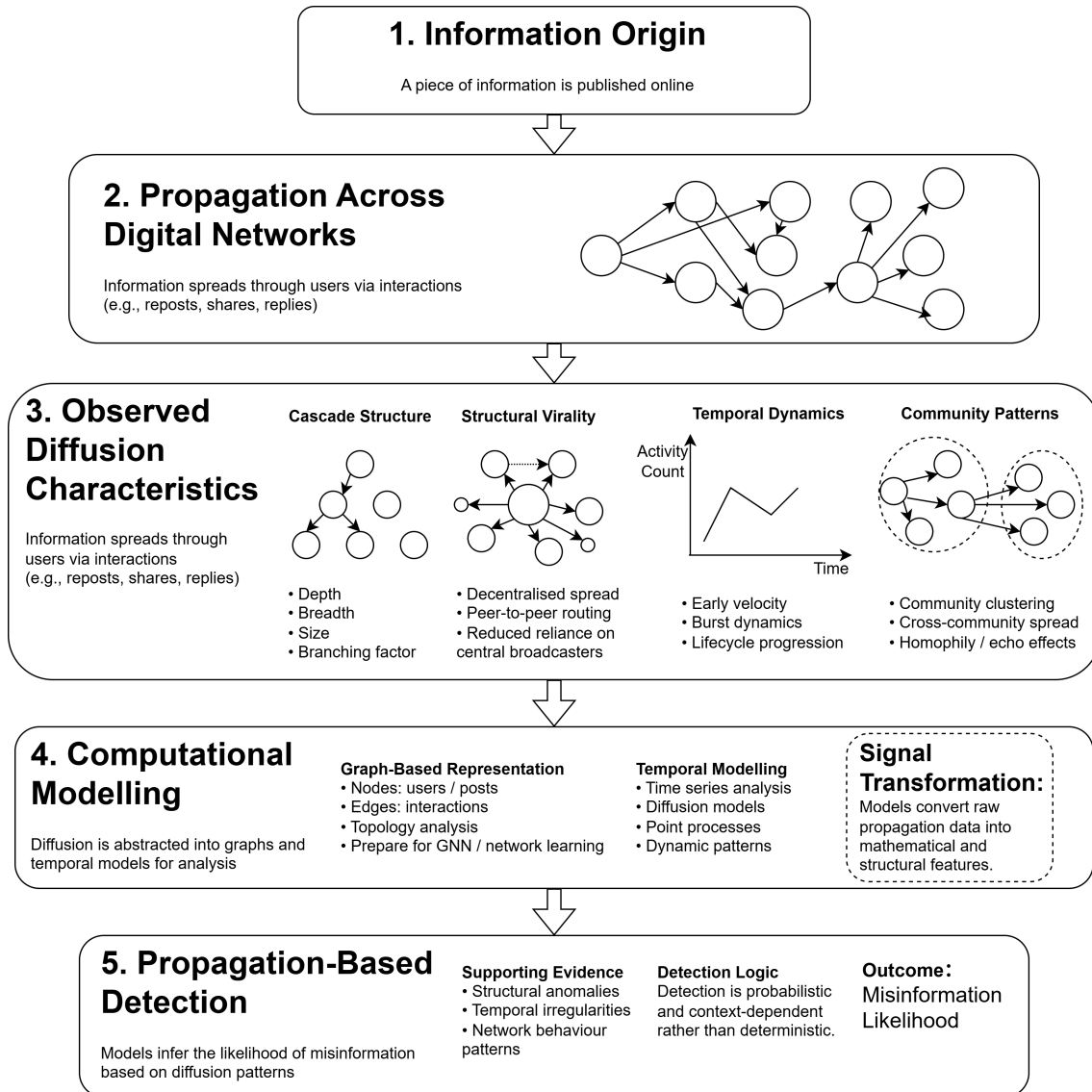


Figure 6.1: Conceptual operational process of propagation-based misinformation detection

only by internal network propagation, but also by external events and time-varying contextual influences. Collectively, these studies suggest that temporal propagation patterns may function as probabilistic indicators of misinformation diffusion.

As computational methods advanced, propagation-based detection increasingly incorporated graph-based learning approaches capable of modelling complex network structures. These approaches represent information diffusion as graphs composed of nodes and edges,⁷ enabling computational models to analyse community structures, propagation pathways, network topology, and relational interaction patterns. More recently, graph neural networks (GNNs)⁸ have become increasingly important within misinformation detection research. Unlike earlier deep learning approaches primarily focused on textual semantics, graph-based learning models are designed to capture relational and structural properties within networked communication systems. For example, [Monti et al. \(2019\)](#) demonstrate how geometric deep learning and graph-based architectures can improve fake news detection by modelling social diffusion structures and user interaction networks directly. These developments illustrate a broader shift towards analysing misinformation not simply as isolated textual content, but as a relational phenomenon embedded within networked communication systems.

At the same time, researchers increasingly recognised that diffusion dynamics alone may not fully explain misinformation propagation. This led to the emergence of hybrid detection approaches integrating content analysis, propagation behaviour, user interaction, and temporal information within unified detection frameworks. For example, [Ruchansky et al. \(2017\)](#) propose the CSI model, which combines textual, temporal, and user-related features for misinformation detection. Similarly, [Shu et al. \(2019\)](#) emphasise the importance of incorporating social context into misinformation detection systems, arguing that information credibility is often shaped not only by content itself, but also by surrounding interaction patterns and network environments. These hybrid approaches reflect broader attempts to capture the multi-dimensional nature of misinformation diffusion within social media ecosystems.

Despite their usefulness, propagation-based approaches face several important limitations. A key challenge is that unusual diffusion patterns do not necessarily indicate misinformation. Legitimate news, breaking events, and emotionally significant information can also spread rapidly and generate large cascades across communication networks ([Vosoughi et al. 2018b](#)). Moreover, diffusion behaviour is strongly influenced by context, including platform architectures, recommendation systems, user behaviour, and external events. As a result, propagation patterns alone cannot provide definitive evidence of misinformation.

These challenges become even more significant in adaptive communication environments. As discussed throughout previous chapters, misinformation diffusion emerges through ongoing interactions between users, platforms, governance mechanisms, and changing social conditions. Detection systems may therefore struggle when diffusion patterns evolve or are intentionally manipulated. Propagation-based approaches provide valuable insights into the structural and temporal dimensions of misinformation, but the signals they identify remain probabilistic indicators rather than definitive markers of misinformation.

⁷In graph-based misinformation detection, nodes typically represent users or pieces of information, while edges represent social connections or information-sharing relationships between them.

⁸Graph neural networks (GNNs) are deep learning models designed to analyse graph-structured data by learning relationships and dependencies between interconnected nodes.

6.3 Behavioural and User-Centred Detection Approaches

As misinformation research expanded beyond content and diffusion analysis, increasing attention was directed towards the behaviour of users and coordinated actors operating within digital communication environments. Behavioural and user-centred detection approaches are based on the assumption that misinformation activity may generate identifiable behavioural signals through patterns of interaction, coordination, amplification, and engagement. Rather than focusing on what information says or how it spreads, these approaches examine how users behave within online platforms, using behavioural indicators such as posting activity, interaction timing, coordinated behaviour, and engagement patterns to identify potentially misleading or manipulative activity.

One major area of behavioural detection research focuses on social bots.⁹ Early research demonstrated that automated accounts can significantly influence online communication environments by rapidly amplifying information, increasing artificial engagement, and contributing to large-scale information manipulation campaigns. Reviewing the broader emergence of automated misinformation activity, [Ferrara et al. \(2016\)](#) highlight how social bots increasingly shape online political communication and information diffusion. As a result, misinformation detection systems began incorporating behavioural indicators such as posting frequency, interaction timing, reposting behaviour, follower-following ratios, and activity regularity in order to distinguish automated accounts from ordinary users.

As behavioural analysis techniques became more sophisticated, researchers developed computational systems specifically designed to identify bot-like activity patterns. One influential example is Botometer,¹⁰ introduced by [Varol et al. \(2017\)](#), which analyses multiple behavioural and interaction features to estimate the probability that an account behaves as a social bot. Such approaches operationalise behavioural detection by transforming user activity patterns into measurable computational signals for misinformation-related risk assessment.

Beyond individual bots, researchers increasingly recognised that misinformation campaigns often involve coordinated behaviour across groups of accounts rather than isolated automated actors. This led to the emergence of coordinated behaviour detection approaches focused on identifying synchronised interaction patterns, coordinated reposting, and organised amplification campaigns.¹¹ Analysing alternative media narratives surrounding mass shooting events on Twitter, [Starbird \(2017\)](#) demonstrate how networked communities collaboratively reinforce and amplify misleading narratives through coordinated online interaction. Similarly, [Pacheco et al. \(2021\)](#) develop computational approaches for identifying coordinated networks on social media by analysing temporal synchronisation, co-sharing behaviour, and interaction similarity patterns between accounts. These approaches reflect a broader shift towards detecting misinformation through collective behavioural structures rather than through isolated pieces of content alone.

Another important direction within behavioural detection involves engagement

⁹Social bots are automated or semi-automated accounts that interact within social media platforms and may imitate human behaviour while disseminating or amplifying information.

¹⁰Botometer is a machine learning system designed to estimate the likelihood that a social media account exhibits automated bot-like behaviour.

¹¹For example, a group of accounts may repeatedly share the same message, hashtag, or URL within a short period of time in order to artificially increase its visibility and perceived popularity.

anomaly analysis. Such approaches examine unusual patterns of interaction including abnormal repost frequencies, artificial engagement spikes, rapid amplification, and irregular temporal activity. The underlying assumption is that misinformation campaigns may generate behavioural patterns that deviate from ordinary communication dynamics. For example, [Grinberg et al. \(2019\)](#) demonstrate that misinformation diffusion during the 2016 United States presidential election was disproportionately concentrated among relatively small groups of highly active users. These findings contributed to the development of anomaly detection systems designed to identify suspicious amplification behaviour within large-scale online communication environments.

Behavioural detection approaches also increasingly incorporate credibility and reputation-based systems.¹² Rather than analysing misinformation directly, these approaches evaluate the trustworthiness of users, accounts, or information sources. Such systems may rely on credibility scoring, reputation estimation, interaction histories, or network trust relationships in order to assess the likelihood that particular actors contribute to misinformation diffusion. Similarly, broader misinformation detection systems increasingly integrate behavioural, social, and contextual information alongside textual analysis. For example, [Shu et al. \(2019\)](#) emphasise that misinformation detection benefits from incorporating social context and interaction patterns rather than relying exclusively on content analysis alone. These developments illustrate how behavioural detection approaches increasingly attempt to capture the broader social environments within which misinformation circulates.

Taken together, these approaches illustrate the diverse behavioural paradigms through which misinformation-related activity is operationalised within behavioural detection systems. Table 6.2 summarises the major forms of behavioural indicators, detection objectives, and associated limitations discussed throughout this section.

As summarised in Table 6.2, behavioural detection approaches each rely on different forms of behavioural inference and consequently face distinct methodological and interpretive limitations. Despite these advances, behavioural and user-centred detection approaches also face broader challenges regarding behavioural ambiguity, contextual variation, adaptive behaviour, and governance concerns.

One major challenge is that suspicious or coordinated behaviour does not necessarily indicate misinformation activity. Legitimate political activism, fandom communities, social movements, emergency communication, and highly engaged online communities may also exhibit coordinated or high-frequency interaction patterns similar to those observed in misinformation campaigns ([Starbird 2017](#), [Marwick & Lewis 2017](#)). Consequently, behavioural signals alone cannot serve as definitive indicators of misinformation.

In addition, behavioural norms vary substantially across platforms, cultures, political environments, and online communities. User interaction patterns that appear suspicious in one context may represent ordinary communication behaviour in another. As discussed in previous chapters, online behaviour is also shaped by recommendation systems, engagement optimisation mechanisms, moderation policies, and platform governance structures. Behavioural patterns therefore cannot be fully separated from the socio-technical environments within which they emerge ([Gillespie 2018](#)).

Another significant limitation concerns adaptive behaviour. As detection systems

¹²For example, detection systems may assign different credibility scores to accounts based on factors such as account age, posting history, interaction behaviour, and previous involvement in misinformation-related activities.

Table 6.2: Major behavioural detection approaches in misinformation detection systems

Behavioural Approach	Behavioural Indicators Analysed	Detection Objective	Key Limitations
Bot detection	Posting frequency, automated interaction timing, follower-following ratios, repetitive activity patterns	Identify automated amplification behaviour and bot-like activity	Human-like bot adaptation and behavioural imitation
Coordinated behaviour detection	Coordinated reposting, co-sharing behaviour, interaction similarity, temporal synchronisation	Detect organised amplification campaigns and coordinated manipulation	Legitimate collective activity may appear suspicious
Engagement anomaly detection	Abnormal engagement spikes, irregular amplification patterns, unusually concentrated activity	Identify suspicious behavioural irregularities and artificial amplification	Legitimate viral events may resemble anomalies
Credibility and reputation systems	Interaction histories, credibility scores, trust relationships, reputation patterns	Estimate source reliability and misinformation-related risk	Credibility and trust remain socially contextual and politically contested

become more sophisticated, misinformation actors increasingly modify behavioural strategies in order to avoid detection. Automated accounts may imitate human interaction patterns, while coordinated campaigns may deliberately reduce synchronisation visibility or diversify interaction behaviour. This ongoing adaptive relationship between detection systems and misinformation actors creates a recursive dynamic in which behavioural indicators continuously evolve over time ([Cresci et al. 2017](#), [Ferrara 2020](#)).

Behavioural detection approaches also raise important governance and ethical concerns. Systems designed to identify suspicious behaviour may produce false positives,¹³ potentially affecting ordinary users, journalists, activists, or politically marginalised groups. Moreover, behavioural surveillance systems may contribute to broader concerns regarding platform transparency, moderation bias, privacy, and algorithmic governance ([Gorwa 2019](#), [Gillespie 2018](#)). Because behavioural detection systems increasingly influence content visibility, moderation decisions, and platform governance processes, these approaches remain closely intertwined with broader questions of accountability, legitimacy, and political power within digital communication environments.

Behavioural and user-centred detection approaches thus extend misinformation detection beyond content and diffusion analysis by incorporating user interaction, coordination patterns, and amplification behaviour. However, behavioural signals remain partial and context-sensitive representations of broader socio-technical dynamics rather than stable or universally reliable indicators of misinformation activity. Human behaviour within digital communication systems is adaptive, socially situated, and continuously shaped by evolving platform environments, governance structures, and strategic interaction between users and detection systems. As discussed in Chapter 5, misinformation engagement is often influenced by emotional responses,

¹³False positives refer to situations in which legitimate users or behaviours are incorrectly classified as suspicious or misinformation-related activity.

heuristic judgement, group identity, social reinforcement, and context-dependent interpretation processes that cannot be fully reduced to stable behavioural indicators alone. Consequently, behavioural detection approaches remain probabilistic and interpretive rather than definitively objective systems for identifying misinformation.

6.4 From Detection to Adaptive Challenges

The detection approaches discussed in previous sections demonstrate that misinformation can be operationalised through multiple analytical perspectives, including content characteristics, diffusion structures, and behavioural interaction patterns. While these approaches provide valuable insights into different dimensions of misinformation activity, they also reflect different assumptions regarding how misinformation can be identified and measured within digital communication environments. Table 6.3 summarises the primary objectives, commonly analysed data, and practical roles associated with these contemporary detection approaches.

Table 6.3: Summary of contemporary misinformation detection approaches

Approach	Primary Objective	Commonly Analysed Data	Practical Role in Detection Systems
Content-Based Detection	Identify misleading semantic or linguistic characteristics	Textual content, images, multimodal content, semantic representations	Most widely deployed foundational detection approach
Network and Propagation-Based Detection	Identify structural or diffusion abnormalities	Cascade structures, reposting patterns, graph topology, temporal propagation traces	Commonly used for large-scale diffusion monitoring and propagation analysis
Behavioural and User-Centred Detection	Identify suspicious interaction or coordination behaviour	User interaction patterns, synchronised activity, engagement behaviour, account coordination	Frequently used for bot detection, coordinated campaigns, and manipulation monitoring

Although these approaches differ substantially in analytical focus and implementation, they also share a common limitation: each operationalises only partial dimensions of broader misinformation ecosystems (Shu et al. 2017b). Content-based approaches primarily operationalise semantic characteristics, propagation-based approaches operationalise diffusion structures, and behavioural approaches operationalise interaction patterns and coordination dynamics. As a result, misinformation detection often becomes fragmented across multiple analytical layers.

These limitations become increasingly significant within contemporary platform environments. As discussed in the previous chapter, recommendation systems, algorithmic amplification, moderation practices, and platform governance increasingly shape how information becomes visible and circulates online (Gillespie 2018). At the same time, emerging communication environments are increasingly characterised by multimodal communication, AI-generated content, adaptive misinformation strategies, and rapidly evolving platform ecosystems (Wardle & Derakhshan 2017, Wu et al. 2024, Przybyła et al. 2025). Under these conditions, misinformation no longer

exhibits fixed semantic, behavioural, or diffusion-based characteristics, making many assumptions underlying traditional detection paradigms progressively less stable.

Consequently, the challenge of misinformation detection extends beyond improving classification accuracy alone. Detection increasingly involves determining which dimensions of communication environments should be observed and how evolving informational conditions influence what becomes visible as misinformation. Understanding misinformation therefore requires moving beyond isolated detection paradigms towards a broader consideration of how misinformation emerges, evolves, and becomes observable within adaptive socio-technical communication environments.

Chapter 7

Reframing Misinformation Through a Cybernetic Perspective

Previous chapters examined misinformation through multiple analytical perspectives including informational systems, diffusion dynamics, behavioural mechanisms, platform environments, and computational detection approaches. Collectively, these discussions demonstrated increasing efforts not only to understand misinformation but also to construct mechanisms through which misinformation becomes observable and actionable within contemporary digital communication systems.

As communication environments become increasingly adaptive and interconnected, contemporary approaches increasingly rely upon observation to support understanding and intervention. However, maintaining stable relationships between computational representations and evolving real-world conditions becomes progressively challenging. This suggests a broader challenge that extends beyond individual detection methods: under what conditions does misinformation remain understandable and observable within contemporary communication environments?

This chapter revisits this challenge by re-examining how existing approaches construct observation and exploring the limitations that emerge as communication environments continue to evolve. It considers how previous perspectives shape what becomes observable before introducing cybernetic perspectives for reconsidering observation within adaptive communication systems. The chapter finally explores how more adaptive forms of misinformation observation may support early warning under changing communication conditions.

7.1 How Does Misinformation Become Observable

7.1.1 Observing and Understanding Misinformation Across Analytical Perspectives

To better understand how this thesis has approached the observation of misinformation from different analytical perspectives, it is useful to briefly revisit the progression developed across the previous chapters perspective.

Chapter 2 established conceptual foundations by positioning misinformation and disinformation as informational phenomena embedded within broader socio-technical systems rather than isolated categories of false content. In doing so, this chapter defined both the object and boundary of understanding by determining

which informational processes become relevant for analysis.

Chapters 3 to 5 then introduced complementary perspectives for understanding misinformation. Chapter 3 examined information diffusion and influence in social networks and focused on how informational movement reveals broader communication dynamics. Chapter 4 examined human behaviour and cognitive mechanisms and explored how participation, interpretation, and interaction shape informational outcomes. Chapter 5 examined platform dynamics and considered how visibility, interaction conditions, and communication environments influence informational circulation.

These perspectives should not be understood as competing explanations of misinformation but as complementary analytical lenses that shape how communication environments become understandable and what later becomes available for observation.

Figure 7.1 summarises this progression. Conceptual foundations define the object of understanding, analytical perspectives determine what can become observed, and computational approaches transform these observations into operational forms.

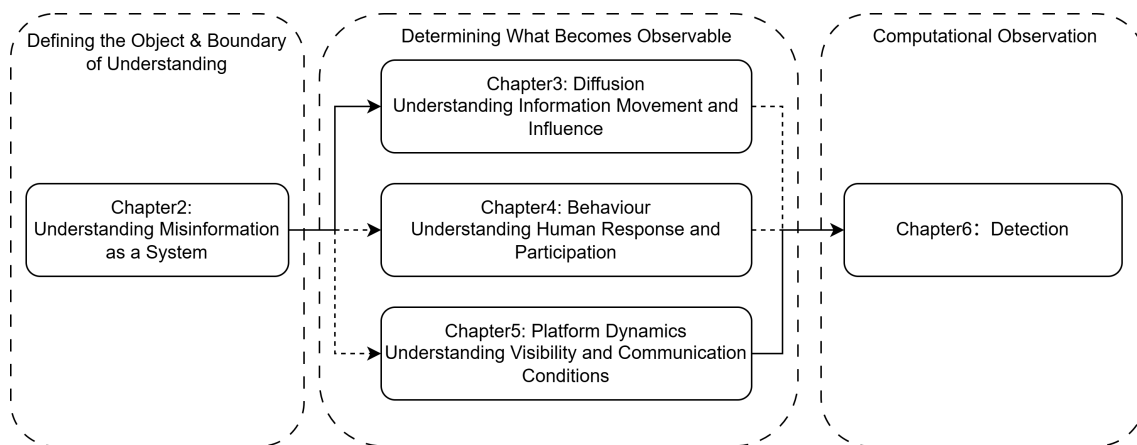


Figure 7.1: Progression across Chapters 2–6. Chapter 2 defines the object and boundary of understanding, Chapters 3–5 provide analytical perspectives that shape what can become observable, and Chapter 6 operationalises observation through computational approaches.

Chapter 6 examined computational detection approaches that organise observable information into measurable representations. Information becomes encoded as features, behavioural processes become represented as observable indicators, and communication environments become incorporated into computational decision structures.

Viewed collectively, Chapters 2 to 6 suggest that contemporary misinformation research increasingly performs understanding through observation and increasingly performs observation through computational mechanisms. Detection therefore emerges not simply as a technical instrument but as an increasingly dominant mechanism through which misinformation becomes understood and acted upon.

At the same time, this progression raises a broader question: If understanding increasingly depends upon observation, and observation increasingly depends upon computational representations and detection mechanisms, under what conditions do these observations remain meaningfully connected to changing communication environments?

7.1.2 Limits of Detection in Maintaining Observation

The previous section demonstrated that contemporary misinformation research increasingly relies on transforming communication environments into observable representations, particularly through computational approaches such as misinformation detection. While Chapter 6 focused on detecting misinformation, this chapter adopts a broader view of observation. Detection is one way of observing misinformation, but observation also involves understanding the conditions under which information becomes visible and meaningful. Here, observation refers not only to identifying misinformation itself, but also to the wider conditions under which information becomes visible, measurable, interpretable, and actionable within communication environments. In this sense, detection may be understood as one computational mechanism of observation rather than observation itself. This distinction becomes increasingly important as communication environments continue to evolve. Detection assumes that observable signals remain sufficiently meaningful for classification and prediction, whereas observation additionally considers how these signals become available, how they are interpreted, and whether they remain connected to changing communication environments.

One important challenge emerges from the instability of the representations upon which computational observation depends. Many detection approaches implicitly assume that labels remain sufficiently stable to support classification and interpretation. Yet unlike many traditional computational settings where labels are treated as relatively persistent representations of underlying categories, misinformation labels are often time-sensitive and context-dependent. Claims may become contested, reframed, partially validated, or reinterpreted as new evidence emerges, scientific understanding develops, political contexts shift, or public interpretation changes.¹ As a result, misinformation increasingly resists being treated as a fixed object with stable observational boundaries. At the same time, the meaning and significance of misleading information often depend not only on informational content itself, but also on surrounding institutional conditions, communication practices, governance responses, and evolving social interpretation. Changes occurring outside computational systems may therefore directly reshape what becomes observable within them.

This challenge has motivated attempts to make computational detection increasingly adaptive under changing conditions. Recent developments including transfer learning, active learning, and continual learning seek to improve model generalisation, adaptive updating, and responsiveness across evolving environments (Yuan et al. 2023, Settles 2009, Parisi et al. 2019). These developments substantially improve the capacity of computational systems to adapt representations under changing conditions. However, adaptation within models does not necessarily preserve meaningful observation. Improving performance across changing datasets may strengthen alignment between models and representations while remaining uncertain whether these representations continue to reflect evolving communication environments.

From this perspective, instability does not necessarily emerge because detection becomes inaccurate. Rather, it emerges because increasingly adaptive environments make it progressively more difficult to maintain meaningful relationships between

¹For example, a message claiming that “the city will introduce water restrictions next month” may initially circulate as misinformation because no official announcement exists. However, if drought conditions worsen and authorities later implement restrictions, the same statement may subsequently become accurate despite no change in wording.

computational representations and the communication environments they seek to describe. Consequently, contemporary misinformation research faces challenges that extend beyond improving detection performance alone. If understanding increasingly depends upon observation and observation increasingly depends upon computational representations, then maintaining meaningful observation itself becomes an increasingly important challenge.

7.1.3 Toward an Observation Gap

The previous section suggested that maintaining meaningful observation becomes increasingly difficult as communication environments continue to evolve. Contemporary misinformation research has substantially expanded the ability to observe and understand misinformation by transforming communication processes into measurable representations. Through content features, behavioural indicators, diffusion structures, and computational modelling, misinformation becomes increasingly analysable and actionable within digital environments.

At the same time, these approaches rely upon an underlying assumption that observable representations remain sufficiently meaningful representations of the communication environments they seek to describe. As communication environments become increasingly adaptive, maintaining this relationship becomes progressively more difficult. Importantly, this challenge does not necessarily emerge because observation immediately fails. Computational systems may continue generating classifications, predictions, and increasingly sophisticated representations under changing conditions. Rather, uncertainty gradually increases regarding whether these representations remain meaningfully connected to the communication environments they were originally intended to describe.

Observation may therefore continue generating increasingly sophisticated representations while becoming less certain whether these representations remain meaningfully connected to changing communication environments².

To describe this condition, this thesis introduces the notion of the **Observation Gap**, defined as the growing uncertainty between computational representations and changing communication environments. Importantly, the Observation Gap does not imply that existing approaches become invalid or that observation becomes impossible. Computational systems may continue improving predictive performance, adapting to changing inputs, and generating increasingly refined outputs while becoming progressively less certain about what these outputs continue to represent outside computational systems.

Under such conditions, improving detection performance alone may not fully resolve broader challenges of misinformation observation. Observation can no longer be understood simply as extracting stable signals from an external reality but increasingly becomes a process through which what becomes visible, measurable, and interpretable is continuously shaped alongside changing communication environments.

This leads to a broader question: if observation increasingly becomes embedded within adaptive communication environments, how should observation itself be

²For example, a weather forecasting system may become increasingly accurate at analysing historical weather records and familiar seasonal patterns. However, if climate conditions gradually change, the system may continue producing highly sophisticated predictions while becoming less certain whether historical relationships still reflect future conditions. Similarly, misinformation systems may continue improving pattern recognition while becoming less certain whether those patterns remain connected to changing real-world environments.

reconsidered under such conditions?

7.2 Reconsidering Observation Through Cybernetic Perspectives

The challenges identified in Section 7.1 suggest that improving misinformation observation may not simply require increasingly adaptive detection systems. While existing approaches continue to expand computational capacity for classification, prediction, and behavioural modelling, these developments do not necessarily resolve the growing uncertainty surrounding whether observable representations remain meaningfully connected to evolving communication environments.

This limitation may indicate a broader issue regarding how observation itself has been understood. Much contemporary misinformation research implicitly treats observation as a process of extracting signals from an external environment and transforming them into increasingly accurate representations. Under such assumptions, improving observation primarily becomes a technical problem of improving data quality, model performance, and computational adaptation.

However, cybernetic perspectives suggest an alternative interpretation.

[Von Foerster \(2003\)](#) shifts attention from observing systems towards observing observing systems, suggesting that observers themselves remain embedded within the systems they attempt to describe rather than occupying an external and neutral position. Later summaries of second-order cybernetics further describe observation as a recursive process in which descriptions and observations continuously interact with the environments they seek to understand ([Heylighen & Joslyn 2001](#)).

From this perspective, observation does not simply identify stable properties that already exist within communication environments. Instead, observation increasingly takes place within adaptive systems where the act of observing contributes to shaping what becomes visible and meaningful. The instability identified in Section 7.1 therefore does not necessarily indicate that misinformation observation has become ineffective. Rather, it suggests that maintaining meaningful observation depends on understanding how different actors participate in shaping the conditions under which information becomes observable.

This perspective also provides an alternative way of understanding human participation within misinformation systems. Existing approaches often observe misinformation through content production, behavioural indicators, interaction structures, and computational prediction, treating human actors primarily as observable entities whose future responses can be estimated. While these approaches remain highly valuable, they may understate an important characteristic of human participation: people are not only sources of data but also observing agents capable of interpretation, reflection, adaptation, and intervention.

This introduces a further challenge. As observation becomes increasingly distributed across automated systems, how can opportunities for human interpretation and intervention continue to be preserved when observations become uncertain? This question shifts attention from improving prediction alone towards understanding how observation itself is organised and maintained within adaptive communication environments.

[Bell \(2020\)](#) raises a related concern in discussions surrounding increasingly automated AI systems by asking where agency remains located under conditions of expanding automation. Rather than treating agency as something displaced by au-

tomation, Bell suggests that maintaining opportunities for agency remains essential for shaping, regulating, and redirecting system behaviour.

Applied to misinformation observation, this perspective suggests that computational systems need not fully replace human participation. Rather than producing final determinations of truth and falsity, computational observation may be better understood as a mechanism for identifying conditions under which human interpretation, reflection, and intervention become necessary. In this sense, computational systems support observation by highlighting moments where existing representations become uncertain, unstable, or potentially disconnected from changing communication environments. Human agency therefore does not replace computational detection but becomes part of a broader adaptive observational process.

Instead of asking how misinformation can be automatically identified under all circumstances, cybernetic perspectives encourage asking how observation can remain adaptive while preserving opportunities for interpretation, intervention, and renewed judgement. This shift moves attention beyond static detection towards forms of observation oriented towards anticipation, reflexivity, and early warning.

7.3 Toward an Early Warning Framework with Human Agency

The previous sections argued that contemporary misinformation observation increasingly faces challenges not simply because detection becomes technically difficult, but because maintaining meaningful relationships between computational representations and changing communication environments becomes increasingly uncertain. Section 7.2 further suggested that observation may need to be reconsidered as an adaptive process involving both computational systems and human agency.

Building on these discussions, this section proposes a conceptual early warning framework that shifts the objective of misinformation observation away from automatically determining whether information should be classified as misinformation and towards identifying when existing observations may require renewed interpretation.

Importantly, recognising the importance of human agency does not imply abandoning computational observation. If computational observation remains valuable but increasingly struggles to maintain alignment with changing communication environments, its role may need to change. Rather than producing final determinations of misinformation, computational observation may instead support identifying conditions under which existing representations become increasingly uncertain and require renewed interpretation.

From this perspective, early warning becomes not an alternative to detection but a mechanism for preserving meaningful relationships between computational representations and changing environments while creating opportunities for human intervention.

Accordingly, the framework treats observation as a continuously adaptive process whose reliability may vary over time and estimates when current representations may no longer remain sufficiently connected to changing communication environments.

To illustrate the proposed framework and clarify the relationships between challenge events, observation dynamics, and human intervention, Figure 7.2 presents the overall conceptual architecture.

Suppose that at time t , the system maintains a current accepted representation:

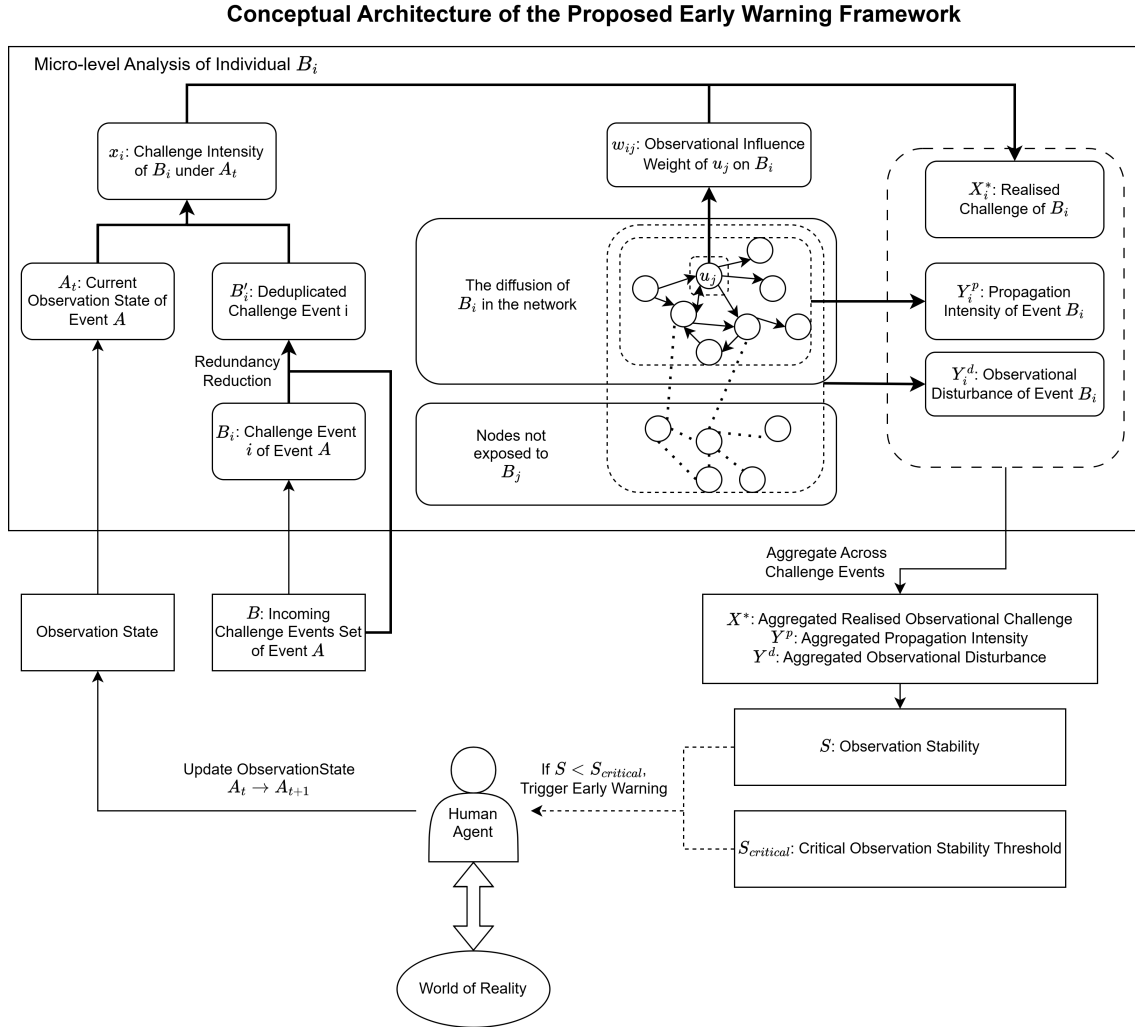


Figure 7.2: Conceptual architecture of the proposed early warning framework for adaptive misinformation observation. Incoming challenge events are transformed into observational challenges through interaction and propagation processes and aggregated into observation-level indicators of challenge, propagation, and disturbance. Observation stability is continuously evaluated, and when stability falls below a critical threshold, early warning activates opportunities for human intervention to update observational states and reconnect computational observation with changing communication environments.

$$A_t$$

which reflects the present observational state constructed through existing information, historical labels, prior interpretations, and previous interventions.

Rather than assuming that observations become unstable through isolated informational events, the framework assumes that multiple informational events may simultaneously challenge existing representations. Let:

$$B = \{B_1, B_2, \dots, B_n\}$$

represent incoming informational events that interact with the current representation.

Because different events may express similar claims, share narratives, or emerge from overlapping sources, the framework first transforms these events into a semantic or narrative representation space:

$$M_B = \phi(B)$$

where M_B represents a matrix of semantic, contextual, or narrative features derived from the incoming events.

A redundancy reduction process is then applied:

$$B' = D(M_B)$$

where:

$$B' = \{B'_1, B'_2, \dots, B'_m\}, \quad m \leq n$$

represents a reduced set of challenge events after removing or merging highly similar, duplicated, or dependent signals. This step prevents multiple versions of the same challenge from being counted repeatedly.

For each reduced challenge event B'_i , the framework estimates an observational challenge:

$$x_i = f(A_t, B'_i)$$

where x_i estimates the degree to which B'_i introduces tension into the current observational state.

Importantly, challenge does not represent factual contradiction alone. Instead, it may reflect semantic divergence, contextual shifts, changes in interpretation, emerging evidence, or inconsistencies with existing observational assumptions.

The collection of these challenge values forms an observational challenge vector:

$$X = (x_1, x_2, \dots, x_m)$$

where each component represents the degree to which a reduced challenge event destabilises the current representation.

However, challenge does not operate only at the content level. Each challenge event also circulates through interaction networks and may be amplified, weakened, or reshaped by different actors.

Let:

$$U = \{u_1, u_2, \dots, u_k\}$$

represent the relevant nodes or actors in the communication environment.

For each challenge event B'_i and node u_j , the framework estimates an observational influence weight:

$$w_{ij} = h(I_{ij}, C_j, P_{ij}, T_j, S_{ij})$$

Importantly, these variables are not intended to represent a fixed or exhaustive specification of observational influence.

Rather, they illustrate several possible dimensions through which challenge events may become amplified, stabilised, suppressed, or transformed as they circulate through communication environments.

At this stage, the framework treats these variables as conceptual placeholders intended to indicate possible mechanisms of observational influence rather than pre-determined operational measures.

Possible dimensions include:

- I_{ij} : interaction intensity between node u_j and challenge event B'_i
- C_j : structural position or centrality of node u_j
- P_{ij} : persistence of challenge event B'_i as it propagates through node u_j
- T_j : contextual trust associated with node u_j
- S_{ij} : temporal stability of interactions surrounding challenge event B'_i

These dimensions are intentionally left open to future operationalisation and may be substituted, extended, or adapted depending on empirical context and observational objectives.

These weights do not represent truthfulness or authority. Rather, they estimate the extent to which particular nodes may amplify, stabilise, or reshape the observational influence of a challenge event.

Importantly, observational challenge and observational influence are treated as operating at different analytical levels.

The challenge value:

$$x_i$$

represents an event-level observational challenge. It estimates the degree to which challenge event B'_i introduces tension into the current observational state A_t , independent of how the event subsequently circulates through communication environments.

In other words, x_i does not indicate whether event B'_i is true or false, nor does it represent individual reactions to the event. Instead, it estimates the potential difficulty of maintaining the current representation when confronted with challenge event B'_i .

However, challenge alone does not necessarily translate into system-level effects.

Whether a challenge meaningfully affects observation depends upon how it becomes realised through interaction, circulation, and interpretation across communication environments.

The influence weight:

$$w_{ij}$$

therefore estimates the extent to which node u_j amplifies, stabilises, suppresses, or reshapes the observational influence of challenge event B'_i .

The realised system-level challenge generated by challenge event B'_i is then expressed as:

$$X_i^* = x_i \sum_j w_{ij}$$

and the realised challenge vector becomes:

$$X^* = (X_1^*, X_2^*, \dots, X_m^*)$$

This formulation reflects the assumption that challenge emerges at the event level but becomes operational only through interaction structures.

Accordingly, the same challenge event may generate substantially different observational effects depending on how it is encountered, interpreted, and circulated across different actors and communication environments.

The framework then considers how challenge events move through the wider communication system. This movement is separated into two related but distinct dimensions: propagation intensity and propagation-induced observational disturbance.

First, propagation intensity captures the extent to which challenge events become distributed across communication environments.

For each challenge event B'_i , propagation intensity may be estimated as:

$$y_i^p = g(E_i, D_i, Q_i)$$

Importantly, these variables are not intended to define a fixed specification of propagation.

Instead, they illustrate possible dimensions through which challenge events may become amplified, sustained, redistributed, or maintained across communication environments.

Possible dimensions include:

- E_i : engagement generated around challenge event B'_i
- D_i : diffusion behaviour associated with challenge event B'_i
- Q_i : cross-boundary propagation across communities, platforms, or interpretive contexts

These dimensions remain intentionally open and may be substituted, extended, or reformulated depending on empirical context and observational objectives.

Propagation intensity is represented as:

$$Y_p = (y_1^p, y_2^p, \dots, y_m^p)$$

where each component estimates one possible expression of how challenge events accumulate and become distributed through communication processes.

Second, propagation may also generate observational disturbance. This refers not simply to how widely a challenge event spreads, but to whether its circulation begins to reshape the conditions under which the system observes, interprets, and responds to information.

For each challenge event B'_i , propagation-induced observational disturbance may be estimated as:

$$y_i^d = r(\Delta A_i, \Delta U_i, \Delta O_i)$$

Importantly, these variables are not intended to define a fixed specification of observational disturbance.

Rather, they illustrate possible dimensions through which communication processes may reshape observational conditions and alter how information becomes visible, interpretable, and actionable.

Possible dimensions include:

- ΔA_i : changes in collective attention generated by challenge event B'_i
- ΔU_i : behavioural adaptation among users or communities
- ΔO_i : shifts in conditions of observability

These dimensions remain intentionally open and may be substituted, extended, or reformulated depending on empirical context and observational objectives.

Observational disturbance is represented as:

$$Y_d = (y_1^d, y_2^d, \dots, y_m^d)$$

where each component estimates one possible expression of how challenge events reorganise attention, interpretation, and observational conditions across communication environments.

This distinction is important because high propagation intensity does not necessarily imply high observational disturbance. A challenge event may spread widely while leaving existing interpretive conditions largely unchanged. Conversely, a moderately diffused event may generate substantial disturbance if it triggers platform response, changes user behaviour, attracts institutional attention, or reshapes how related information is interpreted.

Observational stability is then estimated by combining representational challenge, node-weighted influence, propagation intensity, and propagation-induced observational disturbance:

$$S = F(X^*, Y_p, Y_d)$$

where lower values of S indicate increasing uncertainty regarding whether current computational representations remain meaningfully connected to changing communication environments.

When:

$$S < S_{critical}$$

the framework does not automatically classify information as misinformation. Instead, the system generates an early warning signal indicating that existing observations may require renewed interpretation.

At this stage, human actors function as observing agents who intervene when observational stability becomes uncertain and reconnect computational representations with changing real-world conditions through contextual interpretation, verification, fact checking, and intervention.

Following intervention, the observational state is updated:

$$A_t \rightarrow A_{t+1}$$

Importantly, intervention does not determine future observation once and for all. Instead, updated representations become the starting point for subsequent observations, allowing challenge realisation, communication dynamics, and observational conditions to be re-evaluated under changing environments.

Crucially, human intervention does not function merely as a corrective mechanism within computational systems. Rather, by reintroducing contextual interpretation, situated judgement, and engagement with changing environments, human actors become a bridge through which computational representations reconnect with real-world conditions.

In this sense, adaptation emerges through repeated cycles of observation and intervention rather than through fixed influence structures. Observation therefore becomes not the final stage of decision making but a recursive process through which computational systems and human actors continuously recalibrate relationships between representations and changing communication environments. Updated representations may subsequently reshape future challenge realisation and observational conditions, allowing observation and communication environments to continuously co-evolve over time and preserving opportunities for maintaining meaningful observation under changing conditions.

While the proposed framework remains conceptual rather than operational, its purpose is not to replace existing detection approaches but to illustrate how observation itself may remain adaptive under changing communication environments. From this perspective, the framework serves less as a solution to misinformation detection and more as an attempt to reconsider how meaningful observation may be maintained as communication environments continue to evolve.

7.4 Conclusion and Contributions

Having progressively moved from conceptual discussions of misinformation towards questions of observation, adaptation, and intervention, this final section steps back from the proposed framework and returns to the broader questions that motivated this thesis. Rather than treating the framework as an endpoint, the preceding discussions invite reflection on how misinformation itself becomes understandable and observable under changing conditions.

This thesis set out to address the central research question: *how can misinformation continue to be understood and observed within adaptive socio-technical communication environments?*

To approach this question, two related sub-questions guided the development of the thesis.

The first sub-question asked: *how do existing approaches construct observational perspectives for understanding misinformation?*

In response, Chapters 2 to 5 demonstrated that misinformation is not directly observed as a stable object but becomes visible through multiple observational perspectives. Conceptual distinctions establish boundaries of understanding, diffusion dynamics reveal how information circulates and accumulates influence, behavioural mechanisms explain how actors interpret and respond to information, and platform environments shape conditions of visibility and interaction. Together, these perspectives showed that misinformation observation emerges through multiple interacting

processes rather than through any single analytical viewpoint.

The second sub-question asked: *what challenges emerge when computational observation encounters adaptive communication environments, and how may observation remain meaningful under such conditions?*

In response, Chapters 6 and 7 argued that although computational approaches substantially expand observational capacity, they increasingly encounter uncertainty as communication environments continue to adapt and reshape conditions of interpretation and visibility. To describe this challenge, the thesis introduced the notion of the **Observation Gap** as the growing uncertainty between computational representations and changing communication environments.

The preceding analyses provide a direct answer to the central research question of this thesis:

Misinformation is not directly observable as a fixed object. Instead, it becomes understandable and observable through the ways it is defined, diffused, interpreted, and embedded within communication systems. As communication systems continue to evolve through changing platform dynamics, the conditions through which misinformation becomes observable also change. Therefore, maintaining meaningful observation require greater attention to how computational representations remain connected to changing communication environments.

These conclusions have broader implications for misinformation research. While contemporary research has substantially improved the ability of computational systems to model misinformation, the findings of this thesis suggest that maintaining meaningful observation under changing conditions may be an equally important challenge. This thesis makes two related contributions in response to this challenge.

The first contribution is conceptual. This thesis introduces the notion of the **Observation Gap**, referring to the growing uncertainty that emerges when computational representations become increasingly disconnected from the changing communication environments they are intended to observe. By highlighting this gap, the thesis draws attention to a limitation within existing misinformation research. Contemporary approaches largely focus on improving the performance, adaptability, and sophistication of computational models, yet often leave implicit the question of whether the assumptions and representations underlying those models remain meaningfully connected to changing communication environments. As a result, research may continue refining computational representations while becoming increasingly uncertain whether those representations continue to describe the phenomena they were originally intended to observe. The Observation Gap therefore shifts attention from the optimisation of computational models towards the broader relationship between observation, representation, and changing environments.

The second contribution is methodological. The proposed **Early Warning Framework** offers a cybernetic response to this challenge. Rather than treating adaptation solely as a problem of improving computational models, the framework introduces a mechanism through which observational systems can monitor signs of instability and uncertainty within existing representations. When such conditions emerge, warning signals create opportunities for human intervention, enabling existing assumptions, representations, and ground truth conditions to be re-evaluated and updated. In this framework, human agents are not treated as external validators who periodically correct computational outputs. Instead, they become active participants within an ongoing feedback process through which observational systems

can adapt to changing communication environments. By establishing a recursive relationship between computational observation, human intervention, and evolving real-world conditions, the framework provides a potential bridge between computational models and the environments they seek to describe.

More broadly, these contributions reposition misinformation not simply as a problem of detection, but as an ongoing challenge of maintaining meaningful observation under changing conditions. From this perspective, future misinformation research may depend not only on developing increasingly sophisticated computational models, but also on understanding how observational systems can continually adapt while remaining meaningfully connected to the realities they seek to observe.

7.5 Limitations and Future Directions

Although this thesis has proposed a conceptual perspective and framework for reconsidering misinformation observation, several limitations remain.

First, the proposed framework remains conceptual rather than operational. Variables including challenge realisation, observational disturbance, and observation stability were intentionally left open and should not be interpreted as fixed computational specifications. Their purpose was to provide a conceptual structure for thinking about adaptive observation rather than prescribing specific implementation strategies.

Second, while this thesis examined how communication environments may reshape observation, the proposed framework has not yet been empirically validated. As a result, whether the proposed mechanisms are capable of supporting meaningful observation under real-world conditions remains an open question.

Third, this thesis primarily focused on observation as a socio-technical process and therefore did not fully address broader institutional, governance, and normative questions surrounding intervention. The role of organisational structures, governance arrangements, and different forms of human participation in shaping adaptive observation therefore remains underexplored.

Building on these limitations, several directions for future research emerge. First, the proposed framework should be empirically evaluated to examine whether its core concepts remain meaningful across different communication environments and misinformation contexts. Second, future work may explore how concepts such as observational instability, challenge realisation, and observational disturbance can be operationalised as measurable indicators. Third, the framework creates opportunities for investigating early warning systems that focus not only on detecting misinformation itself, but also on identifying when existing representations, labels, or assumptions may no longer adequately reflect changing conditions. Finally, future research may examine how human intervention, ground-truth updating, and computational adaptation can be combined to support ongoing alignment between observational systems and evolving communication environments. Collectively, these directions shift attention beyond improving detection accuracy towards understanding how meaningful observation can be maintained under conditions of continual change.

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